Label-Limited DINOv3 Adaptation for Photovoltaic Crack Inspection: Recognition, Localization and Acquisition Shift

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Abstract

Photovoltaic electroluminescence inspection requires image review and crack localization under limited annotation and acquisition shift. We compare frozen DINOv3 representations, low-rank adaptation (LoRA), full fine-tuning, CNNs, normal references and local published-architecture ports with explicit label and optimization budgets. One frozen registry retains 1,061 model/inference views in 427 recipe families, yielding 1,683 prediction jobs and 6,212 declared endpoint contrasts. On matching SCDD acquisition-held-source confirmation strata, LoRA has higher image AUROC than the adapted CNN in all six source/budget conditions and all eighteen fitted pairs; positive-image Dice is lower in four of six conditions. Under matched UCF supervision of 86 training and 144 validation masks, LoRA obtains mean confirmation Dice 0.485 versus 0.369 for the full supervised local port, with conditional unadjusted difference interval [0.088, 0.148]. The same comparison transfers to SCDD with lower mean Dice despite higher AUROC. On restricted PVEL, full-budget SCDD LoRA reaches known-crack recall 0.952 while misclassifying an author-normal fraction 0.854 at its source threshold. Reference, ensemble and optimization effects vary by endpoint and cohort; fewer trainable parameters do not guarantee faster inference. All registered contrasts, seeds, native-failure strata and fixed population sensitivities are retained. The results support evaluating recognition, native overlap and operating-point transfer jointly. Intervals condition on fitted systems and are unadjusted; reused development exposure, unequal pretraining/optimization and unverified physical grouping preclude universal architecture superiority or operational reliability claims.

1 Introduction

Electroluminescence (EL) images support photovoltaic inspection by exposing spatial patterns associated with cracks and other defects. A useful inspection system must help decide which images require review and where crack evidence occurs. These objectives require different annotations and can disagree: an image can receive a high crack score while its predicted mask misses a thin crack, and a system can localize part of a crack while flagging many crack-negative images. An evaluation based only on image AUROC leaves these distinctions unresolved.

Large pretrained visual encoders offer a practical way to fit small EL datasets. DINOv3 supplies frozen global and patch representations [7]; low-rank adaptation (LoRA) updates a restricted set of encoder weights [14]. Their usefulness depends on annotation requirements, optimization, acquisition conditions and the downstream task. Different pretraining corpora and different training objectives prevent a causal interpretation of an observed DINOv3–CNN difference as an architecture effect.

Our study asks three questions. First, how do frozen features, LoRA and full fine-tuning behave as training-label availability changes? Second, do image-level gains accompany native-resolution crack recovery and acceptable source-frozen review decisions? Third, does local normal-reference information improve a matched dense predictor across acquisition sources? We retain full supervised and semi-supervised local published-architecture ports, classical feature controls, stronger CNN optimization sensitivities and adverse fitted seeds to test these questions.

Nominal-feature matching is established prior work. PatchCore uses pretrained local features and representative memory [16]; AnomalyDINO develops frozen DINOv2 patch-based anomaly detection [17]; and SuperADD already combines DINOv3 and nominal-feature memory in an industrial workshop-challenge report [18]. We do not claim the first use of DINOv3, EL transfer or local normal memory. DINOv2 is reviewed as prior art and is not executed. Our contribution is the controlled, crack-specific assessment of supervision, native localization, operating decisions

and acquisition transfer, including failures. Published scores on different datasets and ontologies are not used as a performance ranking.

2 Data and evaluation design

2.1 Cohorts, annotations and exposure

Table 1 separates development from the fixed confirmation populations. SCDD contributes 588 audited real images from ARTS, CFVS and CSIR [3]. Its internal split contains 351 training, 120 validation and 117 confirmation images, with 156, 54 and 53 annotated crack positives, respectively. Provided capture-family prefixes define grouping units; physical module identity has not been verified. Whole fields of view are retained. A crack-positive label means an explicit crack annotation in the field, not measured electrical failure or central-cell power loss. Crack-negative images may contain other defects.

Table 1: Study populations and inferential scope. All confirmation populations and exclusions were fixed before predictive access. Development populations support selection and are not independent confirmation.

Cohort	Population	Scope and limitations
SCDD	351 train + 120 validation; 117 confirmation	Three acquisition channels. Matching held-source primary strata contain ARTS 43, CFVS 27 and CSIR 47 images. Provided families are not verified modules.
UCF	430 train + 144 validation; 507 confirmation	Source labels inform hyperparameters. The 507-image cohort is group-disjoint and follows the authors' test-real policy; per-file provenance of the available source subset is unresolved.
PVEL	12,326 scored once; 12,292 primary	Primary population: 943 annotated crack/star positives and 11,349 author-normal negatives. Box truth supports coarse localization, not mask Dice. Capture-family intervals are unavailable.
ELPV supplement	1,968 course-development images	Four grouped outer folds for expert-review labels. The 656 course-test images remain unscored.

UCF training and validation labels support source hyperparameter selection [2]. Their reuse by later SCDD-fitted systems means SCDD-to-UCF 507 is sample-disjoint prediction in a development-exposed dataset, rather than selection on a never-observed dataset. SCDD development also informs study-wide methodological choices. A UCF-only gradient-fit history therefore does not make SCDD unseen to the research process. We retain gradient-fit, validation, upstream hyperparameter and unlabelled-image exposure separately.

The PVEL task restricts positives to explicitly annotated crack/star-crack images and negatives to author-normal images [8]. Unlabelled official-test images and uncertain crack-negative other-defect images do not enter specificity claims. A prospective whole-frame integrity audit reviewed 15,963 images and retained 17 strong within-PVEL near-copy pairs. Their 34 members are excluded from the primary view. The full 12,326-image pool and a broader 11,927-image exclusion view remain sensitivities derived from the same saved predictions. Physical recapture history, all cropped overlaps and pretraining membership remain unknown. These constructed populations do not estimate operational prevalence.

2.2 Annotation budgets and native task definitions

Internal SCDD training budgets use 18, 36, 70 or 351 images, corresponding to approximately 5%, 10%, 20% and 100% of the training pool. The same 120 validation labels are additional supervision. Subsets are nested within seed and use the audited labels for class stratification; they do not measure prospective annotation time. The matched UCF 20% comparison uses 86 training masks and 144 validation masks per fit, all contained within the 574-image development/HPO pool. They are not 230 plus 574 distinct labelled images.

Canonical pixel labels distinguish crack, normal and other intrinsic defects while excluding the specified border, text, padding and measurement-artifact categories. UCF unknown index 128 remains ignored. Pixel-loss eligibility

differs from image validity: ignored regions remain visible in the input. Narrow SCDD crack masks and broader UCF annotation regions can produce different native pixel scores. Acquisition, content, annotation extent and class composition are not separately identified causes in this study.

2.3 Frozen confirmation and uncertainty

After all original fitting, full-method, efficiency and supplementary ELPV schedules completed, a complete source review fixed one registry. It retains 1,061 model/inference-view records in 427 recipe families: 317 three-seed families and 110 deterministic full-budget singletons. These are not 1,061 newly trained models or 427 architectures. Teacher 1 and ensemble inference share training; deterministic reuse and overlapping source cohorts do not create independent replications.

The registry declares 1,683 model/cohort jobs and 6,212 endpoint contrasts. All models, thresholds, source temperatures, preprocessing, memories, score directions, populations, examples and scientific evaluator bytes were fixed before predictive access on 27 October 2026. Predictions and the complete saved-output audit finished on 28 October. No target outcome may select a replacement model, endpoint or example. The complete statistical output and ordered reporting review are now available; no scientific choices were altered using these outcomes.

Image discrimination uses AUROC and average precision (AP), with recall, false-positive rate (FPR) and flagged review fraction at the source-selected threshold. AP follows the average-precision definition. Localization uses whole-cohort pooled native pixel AP, positive-image Dice/IoU and eligible normal-structure FPR. PVEL known-crack recall, author-normal FPR and box-point hit rate retain their narrower meanings. Unsupported endpoints remain not estimable; UCF normal-structure FPR is not replaced with a different negative-field endpoint.

Each declared difference is SECOND family minus FIRST family. Where supported, 1,000 paired resamples of provided acquisition families yield conditional percentile intervals for the mean fitted-pair difference. Seeds 11, 22 and 33 remain individually reported. Intervals are unadjusted across the large comparison family, conditional on these fitted systems, and do not represent training-population or independent-domain uncertainty. Native pooled pixel AP is a whole-cohort point estimate, without pixel pseudoreplication. PVEL receives no invented capture-family interval. The fixed native-failure policy retains crack-area bins at 0.5% and 2%, all positives and all negatives, including empty-denominator nulls.

Figure 1 connects the source-label and selection exposures to the single frozen prediction workflow. Complete aggregate source and cost figures are retained in the separate fourteen-page numerical source companion.

3 Models and controls

Whole grayscale images are replicated to three channels, bicubically letterboxed to 336 pixels and normalized using the corresponding pretrained-model convention. DINOv3-S/16 supplies 384-dimensional global and patch vectors; register tokens are excluded from patch pooling. Image-only systems use frozen DINOv3, LoRA or full fine-tuning, with fully adapted ResNet-18, ResNet-50 and EfficientNet-B0 controls [11, 12]. LoRA uses rank 8 and scaling 8 on query/value projections in the last four transformer blocks. Image heads are linear; pretrained CNN batch-normalization statistics stay fixed while affine parameters remain trainable.

Source-only hyperparameter selection and checkpoint rules precede confirmation. Image systems use AdamW [19], synchronized whole-field D4 views and source-validation AUROC stopping. Source validation also fits temperature scaling [15] and the high-recall image operating point. Stronger image-CNN learning-rate controls remain source optimization sensitivities. Frozen CLS, patch-mean, concatenated, scale and larger-encoder probes, plus HOG/statistics logistic and SVM controls [9, 10], remain separate complete systems with unequal search budgets.

Dense DINOv3 reads depths 3, 6, 9 and 12 with a multiscale decoder and detail skips. The adapted DeepLabv3+/ResNet-50 comparator [13] and DINO systems use image BCE, weighted three-class pixel CE and 0.5 times crack soft-Dice loss. Class weights use inverse-square-root source frequencies. The source checkpoint statistic samples at most 8,192 fixed known transformed pixels per validation image; selected checkpoints are evaluated at native resolution. Matching original mask identities does not match pretraining or optimizer work.

The normal-reference arm uses only selected training masks. A frozen-CLS coverage rule selects at most 256 reference images; each query retrieves five and excludes its entire provided capture family. Normalized patch

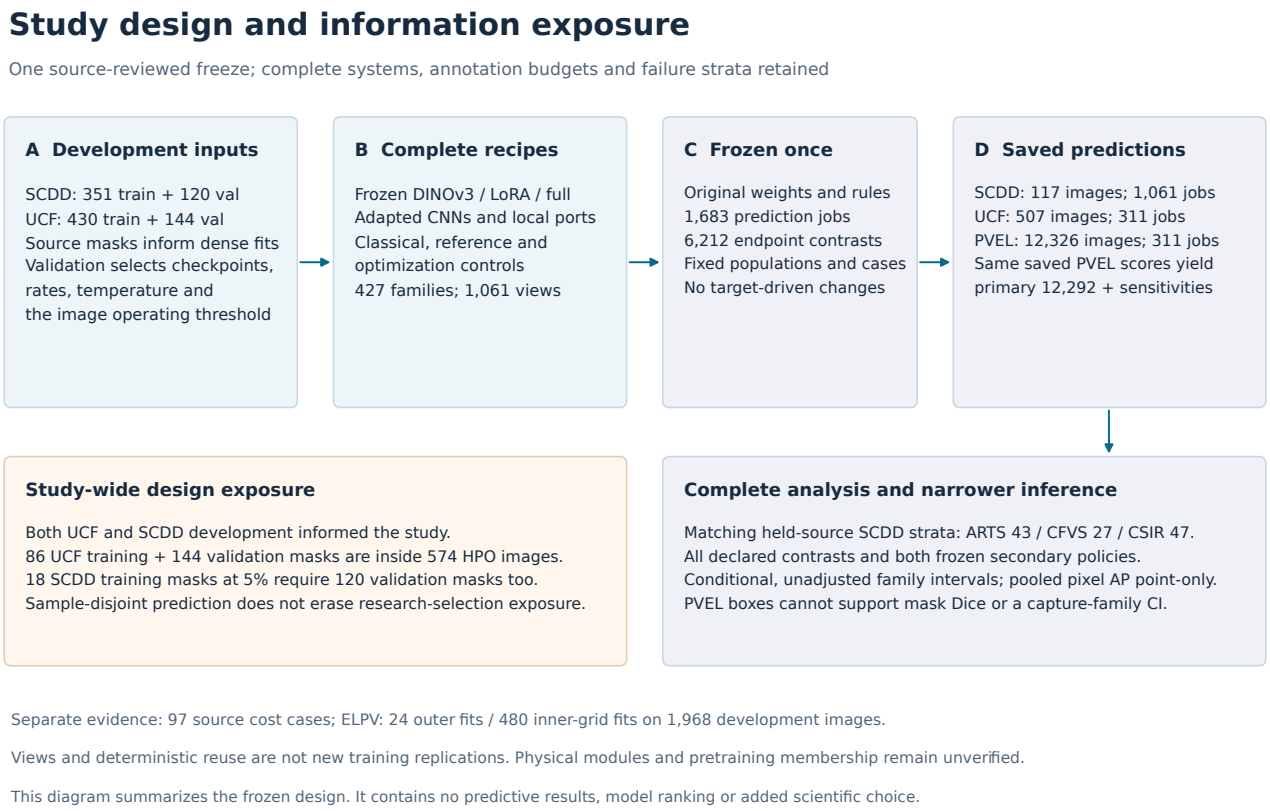


Figure 1: Study design and information exposure. Development labels, hyperparameter selection and acquisition-held-out development can inform study-wide decisions even when confirmation image identities are disjoint. All source-selected recipe families, jobs, contrasts, populations and examples precede predictive access. The diagram reports no performance outcome; family/view counts do not count new architectures or independent training replications.

matching minimizes

$$1 - q^{\mathsf{T}} r + 0.1 \|p_q - p_r\|_2^2. \tag{1}$$

The local input contains query features, query-minus-reference residuals, coordinates and cosine distance. The image head fuses global vectors with top 5% local crack evidence. Reference-free, no-position and mean-evidence controls retain the relevant architectural constraints. Both frozen reference conditions contain 1,387,269 trainable parameters; the LoRA encoder adds 49,152 parameters. A separate capped normal-distance control uses its frozen source noncrack cutoff and is not a calibrated crack posterior. Its cutoff differs from learned softmax threshold 0.5.

Additional controls use a last-layer patch MLP and a frozen post-LoRA encoder with newly trained reference/no-reference decoder heads. Extra adaptation and fresh-head stages are charged as computation; matching masks does not isolate a matching-specific cause. The full supervised and semi-supervised ports retain the released PS-MT/PV-S3 architecture [5, 4], with separately documented three-class whole-field adaptation, teacher-decoder VAT, weak/strong views, CutMix and EMA consistency [22, 21, 20]. Teacher 1 is primary; the fixed teacher-ensemble view is correlated. A literal numerical VAT sensitivity is retained. These are local ports under our protocol, not reproductions of the published headline score. Complete recipes, source-fit counts, source failures and attribution appear in the supplementary record.

3.1 Optimization work and source exposure

Image-only neural fits use batches of eight; joint dense, patch, fresh-head and full-port fits use batches of two. The image and joint dense pipelines use AdamW with weight decay 10^{-4}, 50 updates per round, at most 40 rounds and patience six. The full architecture ports use SGD with momentum 0.9, weight decay 10^{-4} and polynomial learning-rate decay. Their round length is $\max(50, \lceil N_{\text{source}}/2 \rceil)$ updates, using the entire source-training pool size even for a partially labelled supervised subset. Table 2 reports actual saved update counts. These are unequal optimization budgets, rather than matched-compute tests of backbones.

Table 2: Recorded optimization work across the source-run records retained in the frozen catalogue. Ranges span completed rounds, including patience after the selected checkpoint. Update counts are neither FLOPs nor active training time. Correlated teacher/ensemble views share a fit; a fresh-head row excludes its already trained parent stage.

Training pipeline	Batch	Updates/round	Completed updates
Neural image predictors	8	50	350–2,000
Joint dense, patch and fresh heads	2	50	350–2,000
Full supervised/semi-supervised ports	2	124–215	1,000–8,600

The complete recipe ledger retains all 1,061 model/view records and their original 427 family assignments, using 1,043 distinct source-run metadata files and nine source protocols. It records per-seed configurations, actual updates and training/validation/upstream-selection image counts. Recorded label-exposure unions account for overlap, including the matched UCF subset nested within 574 images. They do not measure annotation time, unknown pretraining membership, independent training replication or all upstream grid-search work. Distance and classical predictors have separate memory/solver records, not imputed neural update counts. Original pause-contaminated timers remain raw; no active-compute total is inferred.

4 Source development and computational context

All original source schedules finished before confirmation. The complete source record retains every selected and adverse fitted seed, all ten SCDD dense development conditions, matched UCF controls, full architecture ports, representation/optimization sensitivities and reference mechanisms. Source image AUROC and native crack overlap already disagreed: full-budget internal LoRA had Dice 0.149 versus 0.198 for the adapted CNN despite higher image AUROC. The stronger image-only ResNet-50 optimization sensitivity reached source AUROC 0.969 versus 0.956 for image-only LoRA. Neither source comparison was used to replace a model after confirmation access.

The 97-case formal benchmark contains 388 cohort/batch conditions and 1,164 uninstrumented cohort passes on the 32 GiB M2 Max. Full-budget dense LoRA averages 31.73 ms/image on SCDD at batch one versus 27.08 for the adapted CNN, with mixed fitted-pair timing directions. These cases do not independently benchmark every later confirmation checkpoint. RSS lifetime highwater and sampled MPS tensor/driver allocations remain separate overlapping observations. The separate ELPV development supplement has 24 outer fits and 480 inner-grid fits: DINOv3 CLS plus RBF SVM has mean outer AUROC 0.904918 versus 0.838261 for HOG/statistics plus RBF SVM. Overlapping grouped folds and expert-review labels do not provide crack-localization or independent CNN confirmation. The 656 course-test images remain unscored. Complete source values and figures are preserved in the supplementary record.

5 Confirmation results

All 1,683 frozen prediction jobs, their saved-output audits and all 6,212 declared endpoint contrasts are complete. Every declared contrast is estimable in its registered scope; this does not make every per-job endpoint available. UCF normal-structure FPR remains null, and 116 of the 2,480 native-failure subgroups have empty denominators and null rates. The complete reporting record contains all 16,140 original fitted-pair records, 3,686 absolute family/endpoint cells and all 29 overlapping control-question themes. Tables below summarize original central controls; the exhaustive atlas and numeric companions retain every other contrast, interval, direction and sensitivity. No outcome-dependent replacement or new contrast was introduced.

5.1 Limited-label recognition gains do not ensure native localization

Table 3 shows all four internal SCDD label budgets on the same 117-image confirmation cohort. LoRA has higher mean image AUROC than the adapted CNN at every budget. Positive-image Dice improves at 5%, decreases at 10% and 100%, and differs little at 20%. Pooled native AP favors the CNN at the three larger budgets. The 5% Dice gain also accompanies increased normal-structure FPR, from 0.00112 to 0.00383. Thus the smallest mask budget does not identify an inspection system with uniformly improved discrimination, overlap and false-positive behavior. The 120 validation masks and upstream UCF selection labels remain additional supervision.

Table 3: Complete internal SCDD confirmation controls. Entries are arithmetic means of original per-fit metrics over three fitted seeds, with no pooled model logits. CNN is the adapted joint decoder comparator. Native AP is a whole-cohort point estimate.

Training masks	Image AUROC		Positive Dice		Native AP	
	CNN	LoRA	CNN	LoRA	CNN	LoRA
5%	0.583	0.795	0.052	0.145	0.137	0.151
10%	0.709	0.892	0.196	0.129	0.224	0.176
20%	0.777	0.929	0.163	0.169	0.198	0.162
100%	0.771	0.963	0.181	0.154	0.216	0.168

5.2 Acquisition-held-source recognition and overlap separate

Across all six held-source/budget conditions, LoRA has higher mean image AUROC than the adapted CNN, with positive directions in all eighteen fitted pairs. The conditional unadjusted AUROC interval contains zero for CSIR 20%; the other five intervals lie above zero. These intervals remain part of a 6,212-comparison family and do not imply adjusted significance or independent-domain replication. The small CFVS stratum contains only fifteen positives and twelve negatives; perfect observed AUROC in both budgets does not establish population-perfect prediction.

Dice is lower for LoRA in four of the six conditions (Table 5). CSIR 20% retains lower Dice in all three fitted pairs despite higher AUROC in all three. CSIR 100% instead has higher Dice in every pair, while CFVS 20% has a higher mean with a negative fitted-pair direction retained. Thresholded review decisions also differ: CSIR 20% LoRA increases review from 0.695 to 0.801 and image FPR from 0.420 to 0.638. Matching acquisition strata, rather than pooling uncalibrated models across channels, is essential to these interpretations.

Table 4: All six matching held-source primary contrasts: AUROC. Values are per-fit means; differences are LoRA minus CNN in raw ratio units. Intervals condition on the fitted systems and resample provided acquisition families; all are unadjusted. Signs follow seeds 11, 22 and 33.

Source	Masks	CNN	LoRA	Difference	95% interval	Signs
ARTS	20%	0.556	0.807	0.251	[0.137, 0.339]	+,+,+
ARTS	100%	0.616	0.857	0.241	[0.064, 0.370]	+,+,+
CFVS	20%	0.593	1.000	0.407	[0.284, 0.543]	+,+,+
CFVS	100%	0.894	1.000	0.106	[0.001, 0.238]	+,+,+
CSIR	20%	0.878	0.932	0.054	[-0.029, 0.169]	+,+,+
CSIR	100%	0.624	0.960	0.336	[0.168, 0.551]	+,+,+

Table 5: All six matching held-source primary contrasts: Dice. Values are per-fit means; differences are LoRA minus CNN in raw ratio units. Intervals condition on the fitted systems and resample provided acquisition families; all are unadjusted. Signs follow seeds 11, 22 and 33.

Source	Masks	CNN	LoRA	Difference	95% interval	Signs
ARTS	20%	0.101	0.092	-0.009	[-0.057, 0.015]	-,+,-
ARTS	100%	0.134	0.101	-0.032	[-0.083, 0.002]	+,-,-
CFVS	20%	0.138	0.205	0.067	[0.015, 0.123]	+,-,+
CFVS	100%	0.231	0.205	-0.026	[-0.081, 0.026]	-,+,-
CSIR	20%	0.273	0.179	-0.094	[-0.198, 0.042]	-,-,-
CSIR	100%	0.157	0.189	0.032	[-0.036, 0.120]	+,+,+

Figure 2 includes every original family-pair comparison with both AUROC and Dice declared in the same cohort/stratum: 313 joined pairs in total. Contrasts occur in both agreement and disagreement quadrants. Shared fits, overlapping source populations and paired endpoints prevent treating these points as independent observations or fitting a new confirmatory association.

5.3 Matched UCF supervision helps within cohort but transfers unevenly

The matched 20% UCF comparison retains all four systems trained with the same 86 masks and 144 validation masks (Table 6). LoRA versus the full supervised local port has Dice difference 0.116 [0.088, 0.148] and AUROC difference 0.048 [0.017, 0.085], with conditional unadjusted intervals. Mean recall is the same to displayed precision, while review decreases from 0.797 to 0.748. These results concern this matched annotation setting; they do not match HPO, pretraining or optimizer work.

Table 6: All four matched UCF 20% systems on UCF 507. Ratios are descriptive fitted-seed means. Validation/HPO exposures overlap within 574 development images; UCF normal-structure FPR is unavailable.

System	AUROC	Native AP	Positive Dice	Recall	Review
Frozen DINO	0.915	0.544	0.450	0.978	0.782
LoRA	0.946	0.641	0.485	0.972	0.748
Normal reference	0.900	0.556	0.440	0.971	0.810
Full supervised port	0.898	0.595	0.369	0.972	0.797

The same matched LoRA and supervised port transfer to SCDD 117 with mean AUROC 0.875 versus 0.604 but Dice 0.078 versus 0.096; the Dice difference is -0.018 [-0.040, 0.002]. Thus a favorable UCF Dice comparison does not persist as a uniform cross-cohort gain. At full UCF supervision, the original LoRA system also has slightly higher UCF AUROC than the adapted CNN (0.962 versus 0.951), but lower Dice (0.472 versus 0.492) and native AP (0.635 versus 0.665). Normal reference has lower mean Dice than the matched frozen predictor on both UCF 507 and SCDD 117. The broader reference and fresh-head questions retain both positive and adverse directions, so no matching-specific mechanism is identified.

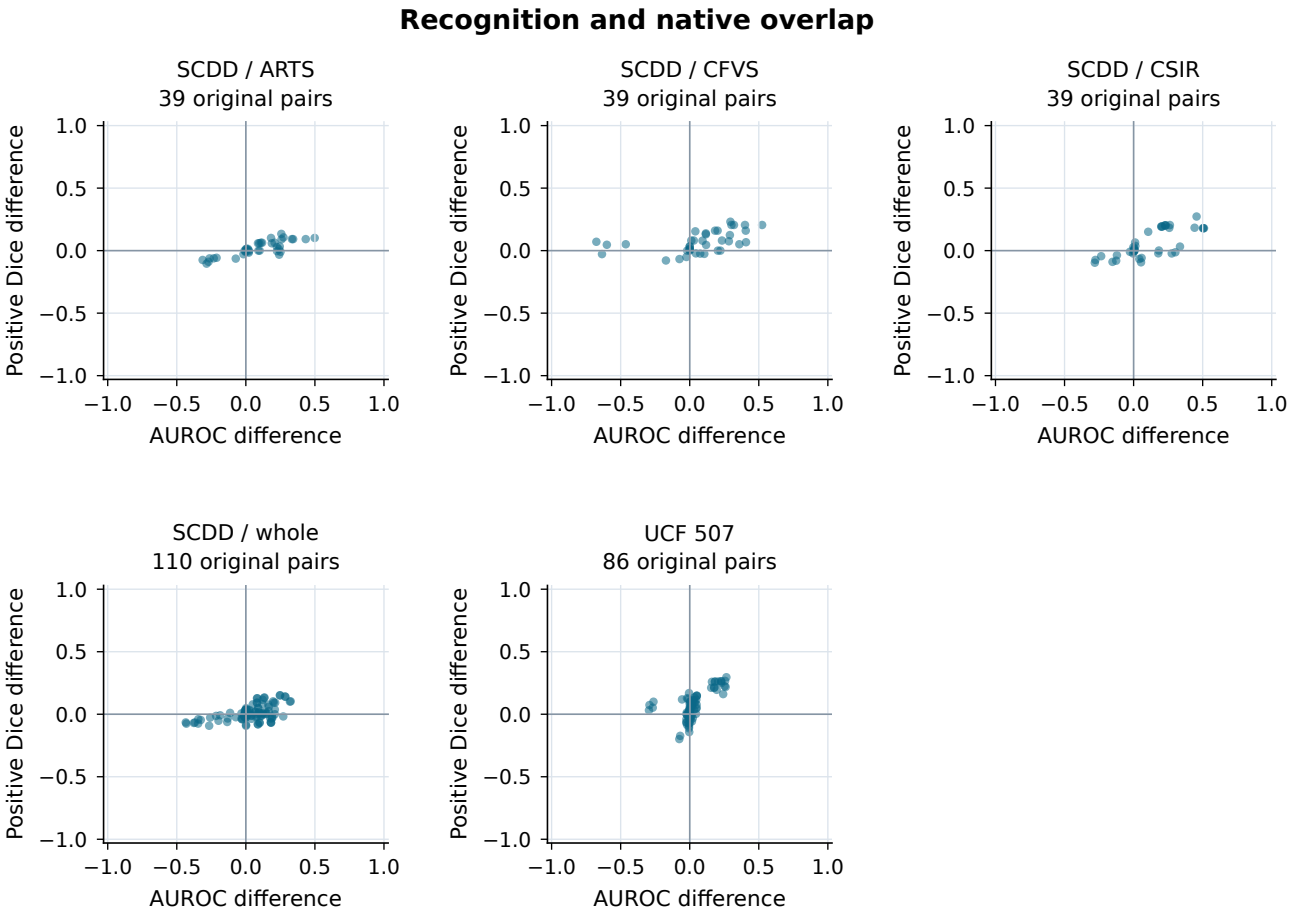


Figure 2: All 313 existing recognition/localization endpoint joins, with literal SECOND minus FIRST differences. Every declared eligible family pair is retained within its original cohort/stratum. Positive AUROC differences can coexist with negative Dice differences. These descriptive, correlated points do not provide a new correlation test or architecture ranking; original conditional unadjusted intervals remain in the complete contrast atlas.

5.4 Source operating points fail under restricted PVEL transfer

The complete PVEL population analysis reuses the same once-scored 12,326-image pool. Table 7 retains all three original source regimes in the dense-control presentation scope, without target-threshold fitting. High known-crack recall coexists with very high author-normal FPR. For example, full-budget SCDD LoRA has recall 0.952 and FPR 0.854; the matched UCF 20% LoRA has recall 0.894 and FPR 0.741. These are failures of portable source operating points in the restricted population, rather than evidence of a deployable high-recall detector.

Table 7: All original source regimes in the PVEL dense-control scope, primary 12,292-image population. Each cell is CNN/LoRA fitted-seed means. Box-point hit is not mask segmentation. No acquisition-family interval is asserted.

Source regime	Known-crack recall	Author-normal FPR	Box-point hit
SCDD 20%	1.000/0.996	1.000/0.981	0.247/0.363
SCDD 100%	1.000/0.952	1.000/0.854	0.542/0.359
UCF 100%	1.000/0.945	0.999/0.848	0.345/0.247

Box localization is not uniformly better for LoRA: box-point hit increases from 0.247 to 0.363 for SCDD 20%, but decreases from 0.542 to 0.359 for SCDD 100% and from 0.345 to 0.247 for UCF 100%. All full-pool, broader-screen and line-only sensitivity records remain reported for all 311 views. Their small or large changes do not validate capture-family independence or remove composition dependence from secondary AUROC/AP.

5.5 Native failures, calibration and secondary controls

Figure 3 retains all 496 native-mask jobs in the fixed crack-area bins. High image recall can coincide with low overlap and frequent literal zero-Dice masks, especially for small annotated crack footprints. Zero overlap need not mean an entirely empty prediction. All-positive and all-negative summaries remain separate; negative-field mean pixel FPR is not substituted for normal-structure FPR. PVEL box annotations never enter mask-Dice panels.

Source-temperature diagnostics retain Brier score, negative log-likelihood and fixed-bin ECE for every original scored model/cohort pool. Improvements are not universal, and no target temperature is fitted. These whole-pool diagnostics use 117 SCDD and 12,326 PVEL images, rather than matching held-source strata or the 12,292-image primary PVEL population. The full 29-question readout separately preserves frozen-feature/classifier controls, input scale and pooling, larger encoders, stronger CNN rates, full supervised/semi-supervised ports, correlated ensembles, literal numerical sensitivities and post-adaptation reference heads. Benefits in one endpoint do not establish complete-system dominance. The stronger image-only CNN controls also remain informative: all twelve original LoRA-minus-additional-CNN AUROC comparisons on SCDD 117 and UCF 507 are positive, with positive directions in all thirty-six fitted pairs. Against the ResNet-50 control at learning rate 10^{-5}, the SCDD difference is 0.014 [-0.034, 0.066], whereas the UCF difference is 0.073 [0.028, 0.134], with conditional unadjusted intervals. The higher source-validation ResNet-50 ranking therefore does not persist uniformly in these confirmation comparisons. These reused fits and cohorts do not count as twelve independent studies or isolate a backbone effect. The fixed spatial record additionally retains all 3,780 original maps in 216 local sheets; no outcome-selected example is substituted.

6 Discussion

The complete confirmation results retain the development pattern of disagreement between recognition and localization. Higher LoRA AUROC across all six held-source conditions coexists with lower mean Dice in four. Within matched UCF supervision, improvement in both recognition and Dice is substantial relative to the supervised local port, but the same fitted systems transfer with adverse Dice on SCDD. This combination supports a task- and population-specific assessment of complete systems, rather than a single foundation-model ranking.

Source-selected high-recall thresholds are particularly fragile under restricted PVEL transfer. High recall with high author-normal FPR and inconsistent box-point recovery does not provide a useful autonomous inspection policy. The fixed population sensitivities describe the consequences of prespecified integrity and annotation restrictions;

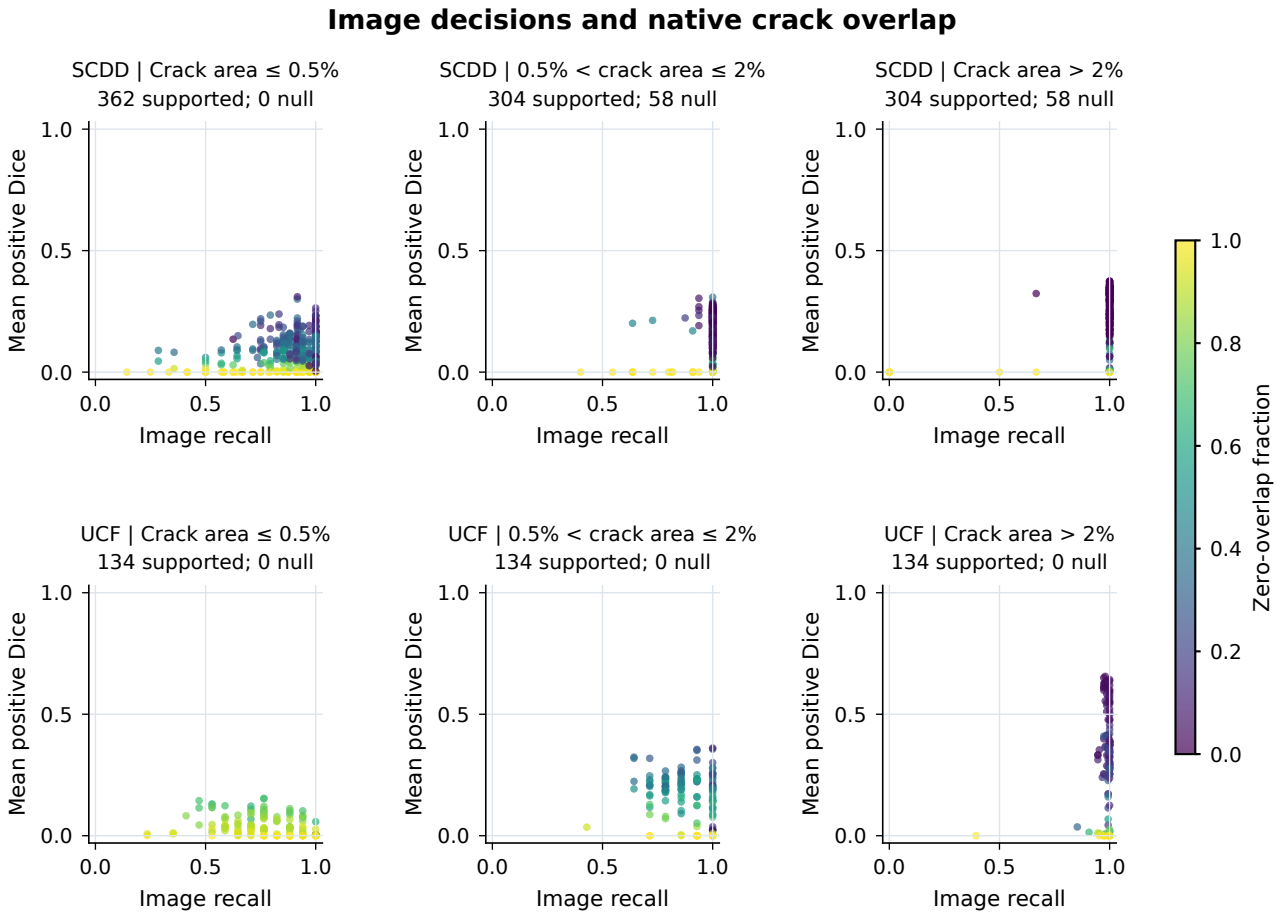


Figure 3: Every fitted native-mask job in its unchanged positive-area strata. Image recall uses the source image threshold; native overlap uses the fixed pixel rule. Crack area is the annotated crack-pixel fraction among known evaluation pixels. Color represents the literal zero-overlap fraction. Empty strata retain null estimates. Fitted records are correlated and these descriptive subgroup plots have no subgroup confidence intervals.

they do not estimate field prevalence or repair a poorly transferred operating point. No target calibration, threshold adjustment or favorable example selection is used to remove these failures.

Normal-reference and fresh-head controls retain mixed effects. Source memory coverage, retrieval, decoder optimization, annotation extent and acquisition appearance remain competing explanations. Equal decoder parameter counts remove one difference but do not establish a matching-specific cause. Similarly, published-architecture and semi-supervised ports differ in optimization work and unlabelled exposure. Correlated teacher/ensemble views and deterministic predictor reuse do not create new training replications.

Several limitations constrain inference. Both UCF and SCDD development informed study-wide design; disjoint confirmation identities do not make either dataset wholly unseen to research selection. Pretraining corpora, objectives and HPO budgets prevent architecture-only attribution. Provided acquisition families do not verify module identity; image integrity screening does not exclude all crop overlaps or pretraining membership. PVEL author-normal specificity does not extend to every other defect, and box truth cannot validate segmentation. UCF source-file provenance remains unresolved. Label-stratified subsets and reused validation masks do not identify annotation labour savings. Finally, the large family of unadjusted conditional intervals does not support multiplicity-adjusted superiority claims.

7 Conclusion

DINOv3 adaptation can improve label-limited crack recognition and selected native-mask comparisons, while localization, review burden and transferred operating points remain separate constraints. Complete fixed-cohort results expose both gains and failures: consistent held-source AUROC directions, mixed Dice effects, a favorable matched UCF mask comparison and high restricted-PVEL false-positive rates. Inspection studies should report these endpoints together, retain weak seeds and unavailable annotations, and make supervision and optimization exposure explicit.

8 Reproducibility and availability

The owner-side record retains immutable protocols, the single registry, ordered data identities, source exposure, original failures/pauses, optimizer/RNG state, selected weights, predictions and raw histories. A registered rerun of the unchanged statistical workflow reproduced all 1,683 saved-job records and 6,212 contrasts exactly apart from generation time, without overwriting the original analysis. This establishes repeatability on the same host, environment, implementation and saved outcomes; it does not establish independent-implementation, cross-device or fresh-training reproduction. The authorized-input package retains the exact four-author-file acquisition recipe and mandatory PV-S3 oracle check, while unknown-licensed upstream source remains excluded from distributable materials. The owner-only dependency index binds every saved prediction product and original cache-retirement receipt. Retired per-image AP intermediates remain distinguished from preserved final native summaries, predictions and fixed maps.

Data and model permissions are artifact-specific. ELPV image and code terms differ; PVEL repository and dataset-form conditions remain separate; absence of an explicit raw-data license in an inspected SCDD/UCF tree is not a blanket redistribution permission. DINOv3 and copied MIT-licensed PS-MT material retain their source attribution and applicable terms. Raw EL pixels/masks, prediction arrays and model/optimizer payloads are excluded from the sanitized private handoff. No public release or external submission is made here.

References

[1] S. Deitsch et al. Automatic Classification of Defective Photovoltaic Module Cells in Electroluminescence Images. *Solar Energy*, 2019. https://github.com/zae-bayern/elpv-dataset.

[2] J. Fioresi et al. Automated Defect Detection and Localization in Photovoltaic Cells Using Semantic Segmentation of Electroluminescence Images. *IEEE Journal of Photovoltaics*, 12(1), 2022. https://doi.org/10.1109/JPHOTOV.2021.3131059.

[3] L. Pratt, J. Mattheus, and R. Klein. A benchmark dataset for defect detection and classification in electroluminescence images of PV modules using semantic segmentation. *Systems and Soft Computing*, 5:200048, 2023. https://doi.org/10.1016/j.sasc.2023.200048.

[4] A. Jha, Y. Rawat, and S. Vyas. Advancing automatic photovoltaic defect detection using semi-supervised semantic segmentation of electroluminescence images. *Engineering Applications of Artificial Intelligence*, 160:111790, 2025. https://doi.org/10.1016/j.engappai.2025.111790.

[5] Y. Liu et al. Perturbed and Strict Mean Teachers for Semi-Supervised Semantic Segmentation. *CVPR*, 4258–4267, 2022. https://openaccess.thecvf.com/content/CVPR2022/html/Liu_Perturbed_and_Strict_Mean_Teachers_for_Semi-Supervised_Semantic_Segmentation_CVPR_2022_paper.html.

[6] M. Gharbage et al. Rethinking Transfer Learning for Industrial Inspection: DINOv3 vs. ImageNet Pretraining Across RGB and X-ray Tasks. *CVPR Workshops*, 2026. https://arxiv.org/abs/2605.23472.

[7] O. Siméoni et al. DINOv3. *arXiv:2508.10104*, 2025. https://arxiv.org/abs/2508.10104. Official models: https://github.com/facebookresearch/dinov3.

[8] B. Su, Z. Zhou, and H. Chen. PVEL-AD: A Large-Scale Open-World Dataset for Photovoltaic Cell Anomaly Detection. *IEEE Transactions on Industrial Informatics*, 19(1):404–413, 2023. https://doi.org/10.1109/TII.2022.3162846.

[9] N. Dalal and B. Triggs. Histograms of Oriented Gradients for Human Detection. *CVPR*, 2005. https://doi.org/10.1109/CVPR.2005.177.

[10] C. Cortes and V. Vapnik. Support-vector networks. *Machine Learning*, 20:273–297, 1995. https://doi.org/10.1007/BF00994018.

[11] K. He, X. Zhang, S. Ren, and J. Sun. Deep Residual Learning for Image Recognition. *arXiv:1512.03385*, 2015. https://arxiv.org/abs/1512.03385.

[12] M. Tan and Q. Le. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *ICML*, PMLR 97:6105–6114, 2019. https://proceedings.mlr.press/v97/tan19a.html.

[13] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. *ECCV*, 2018. https://doi.org/10.1007/978-3-030-01234-2_49.

[14] E. J. Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv:2106.09685*, 2021. https://arxiv.org/abs/2106.09685v2.

[15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On Calibration of Modern Neural Networks. *ICML*, PMLR 70:1321–1330, 2017. https://proceedings.mlr.press/v70/guo17a.html.

[16] K. Roth et al. Towards Total Recall in Industrial Anomaly Detection. *CVPR*, 2022. https://arxiv.org/abs/2106.08265v2.

[17] S. Damm, M. Laszkiewicz, J. Lederer, and A. Fischer. AnomalyDINO: Boosting Patch-based Few-shot Anomaly Detection with DINOv2. *WACV*, 2025. https://arxiv.org/abs/2405.14529v3.

[18] L. Roming et al. SuperADD: Training-free Class-agnostic Anomaly Segmentation—CVPR 2026 VAND 4.0 Workshop Challenge Industrial Track. *arXiv:2605.14808v1*, 2026. Workshop-challenge technical report. https://arxiv.org/abs/2605.14808v1.

[19] I. Loshchilov and F. Hutter. Decoupled Weight Decay Regularization. *ICLR*, 2019. https://arxiv.org/abs/1711.05101v3.

[20] A. Tarvainen and H. Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. *Advances in Neural Information Processing Systems*, 30, 2017. https://proceedings.neurips.cc/paper_files/paper/2017/hash/68053af2923e00204c3ca7c6a3150cf7-Abstract.html.

[21] S. Yun et al. CutMix: Regularization Strategy to Train Strong Classifiers with Localizable Features. *ICCV*, 2019. https://arxiv.org/abs/1905.04899v2.

[22] T. Miyato, S.-i. Maeda, M. Koyama, and S. Ishii. Virtual Adversarial Training: A Regularization Method for Supervised and Semi-Supervised Learning. Author manuscript, *arXiv:1704.03976v2*, 2018 revision. https://arxiv.org/abs/1704.03976v2.

[23] F. Milletari, N. Navab, and S.-A. Ahmadi. V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation. *3DV*, 2016. https://doi.org/10.1109/3DV.2016.79.

Supplementary methods and complete confirmation controls

This supplement accompanies the completed single-registry confirmation analysis. It reports the original methods, all 29 overlapping control-question themes and all 456 endpoint comparisons in the six fixed dense-control table scopes. Complete machine-readable companions retain all 6,212 declarations and 16,140 fitted-pair records, including the other controls, original intervals, population sensitivities and unsupported per-job endpoints. No new model, contrast, calibration, population or example is introduced here. Direction counts describe overlapping comparisons, not independent wins or multiplicity-adjusted tests.

A Detailed source methods and exposure

The main article distinguishes gradient-fit labels, validation labels and upstream hyperparameter exposure. The 18 masks at the smallest SCDD training budget require 120 additional validation masks, plus upstream UCF calibration. Matched UCF20% fits use 86 training and 144 validation masks within the same 574-image HPO pool. Pretraining membership is unknown. These budgets describe available annotations, not measured annotation labour.

A.1 Global frozen representations and image-only adaptation

Whole grayscale images are copied into three channels, resized with bicubic letterboxing and normalized with the corresponding ImageNet convention. Input size is initially 336×336; enlarging an image does not recover absent physical detail. DINOv3-S/16 provides a 384-dimensional CLS vector and patch vectors. Register tokens are excluded from patch pooling. Image classifiers receive a per-image, non-affine layer-normalized vector.

The six primary image systems are DINOv3-S frozen, LoRA and full fine-tuning; and fully fine-tuned ResNet-18, ResNet-50 and EfficientNet-B0. Their final image heads are linear. LoRA uses rank 8, scaling 8 and query/value projections in the last four transformer blocks. It changes encoder projections; it is not an operation applied only to a stored embedding vector. CNN pretrained batch-normalization statistics remain fixed, with affine parameters trainable. Different pretraining data prevent a causal Transformer-versus-CNN interpretation.

Source-only calibration evaluates two learning rates per family and three paired seeds on UCF430/144. Primary real-cohort development contains 180 fits: 72 internal budget fits plus 108 acquisition-source-held-out fits at 20% and 100% source-training labels. Batches use synchronized whole-field D4 views, 50 updates per round, at most 40 rounds, AdamW [19] and source-validation AUROC early stopping with patience six. Source validation selects temperature and a high-recall decision threshold; no target threshold is fitted.

Converged Logistic Regression, Linear SVM and RBF SVM controls use HOG plus 42 intensity/gradient statistics (1,806 dimensions) or frozen CLS features. A separate 379-fit representation study compares CLS, image-area-weighted patch means and concatenation; DINOv3-S at 224/336/448; DINOv3-B at 336; and frozen ResNet-18 classifier probes. Full-budget deterministic probes are stored once. Classical and neural search grids have different sizes; equal tuning cost is not claimed.

A.2 Matched dense networks and local normal structure

An adapted DeepLabv3+/ResNet-50 baseline and a four-depth DINOv3 decoder share image and canonical pixel supervision. The DINO decoder reads transformer depths 3, 6, 9 and 12 and includes detail skips. Both use

$$\mathcal{L} = \mathcal{L}_{\text{image BCE}} + \mathcal{L}_{\text{weighted pixel CE}} + 0.5\mathcal{L}_{\text{crack Dice}}. \tag{2}$$

Overlap-based training objectives have precedents including V-Net [23]. Our crack-class soft Dice term uses linear probability/indicator sums, additive-one smoothing and a per-image mean. V-Net uses squared denominator terms in its volumetric objective. The implemented loss combination is therefore specified here rather than presented as an exact V-Net reproduction.

Pixel class weights follow inverse square-root source frequencies. The source checkpoint metric uses up to 8,192 fixed, uniformly selected known transformed pixels per validation image; full native-mask metrics are reported at the selected checkpoint. Validation masks are charged as supervision.

Only selected training masks define normal references. At most 256 reference images are selected by frozen-CLS coverage; a query retrieves five similar images, then matches their eligible normal patches. The complete query capture family is excluded. For normalized patch vectors q and r, matching minimizes

$$1 - q^\top r + \lambda \|p_q - p_r\|_2^2, \qquad \lambda = 0.1. \tag{3}$$

The local network receives the query, query-minus-reference residual, coordinates and cosine distance. Its top 5% local crack evidence is fused with global vectors for image classification. Reference-free, no-position and mean-evidence controls test the mechanism. The registered dense study has 24 UCF calibration and 144 real-cohort development fits; nine UCF-only reference fits preserve a source-only transfer comparison.

A.3 Decoder dimensions and global/local fusion

At a 336-pixel input, each projected transformer depth supplies a $21 \times 21 \times 128$ feature map. The normal-input slot concatenates normalized query features (384), matched residuals (384), coordinates (2) and cosine distance (1): 771 values project to 64 channels. The reference-free arm retains this same network, with residual and distance set to zero. Combining four depth projections and this slot gives 576 channels, fused to 128. Four grayscale-detail upsampling stages use widths 96, 64, 32 and 24 before the three-class output.

The frozen decoder/image head has 1,387,269 trainable parameters in both reference conditions. Rank-eight Q/V adapters in the last four blocks add 49,152 encoder parameters. Thus a reference comparison changes its input information while retaining decoder architecture and parameter count. The image head concatenates the final CLS and fully image-valid patch mean (768 features), applies non-affine LayerNorm, appends scalar top-five-percent local log-odds, and fits a 769-input linear output. Annotation-ignore regions remain visible in the image input; geometric image validity differs from pixel-loss eligibility.

A.4 Explicit patch, distance and frozen post-LoRA controls

A last-layer patch baseline uses a 128-hidden-unit MLP with three outputs. Patch targets max-preserve thin annotated cracks. Its image BCE, weighted patch CE and patch Dice objective differs from full-resolution segmentation, although the original annotated image identities are the same. Its upsampled map cannot supply arbitrary subpatch detail.

A normal-distance baseline compares query patches against at most 64 deterministically indexed normal patches per reference image/view. Nearest cosine distances are exact within that capped memory and exclude the query family. A source-validation noncrack response quantile fixes its pixel operating point. This mask-informed memory is not annotation-free; the resulting response is not a calibrated crack posterior. Native pixel AP is primary, and its cutoff convention is distinguished from learned softmax-score threshold 0.5.

For post-LoRA references, a source-trained adapter is merged into the original encoder, frozen and re-encoded before references are rebuilt. Two fresh decoder heads, with and without references, start identically and use exactly the parent's budgeted source images/masks. Their extra adaptation/head stages add computation, not new annotation identities. Each encoder export verifies merged and freshly reloaded tokens. Cache identities bind the actual merged weights. Exact canonical reference tensors persist; noncanonical D4 training intermediates are reproducible from source inputs and their recorded hashes. The mechanism study registers 36 patch fits/calibrations, 23 distance controls and 66 post-LoRA fresh-head fits. All 125 registered mechanism fits are complete; their complete source results and counterexamples remain in the separate source numerical companion and archived record.

A.5 Comparison with published semi-supervised methods

PV-S3 [4] and its acknowledged PS-MT basis [5] use teacher/student consistency, rather than the adapted supervised DeepLab baseline alone. We retain the complete MIT-licensed PS-MT deep-stem ResNet-50/DeepLabv3+ architecture, whose released encoder/decoder source matches the pinned PV-S3 architecture byte for byte. Each network has 41,762,595 parameters. The student uses the authors' publicly posted ImageNet initialization; EL-task-trained weights are not used because their overlap with confirmation images is unknown. Two teachers initialize independently, following the released role policy.

The implemented semi-supervised arm includes teacher-decoder virtual adversarial feature perturbation, confidence-weighted CE with positive threshold 0.6, weak/strong photometric views, CutMix with Beta(4,4), and EMA teacher updates with coefficient 0.996. The active teacher alternates by optimization round; after every student optimizer step, the active teacher for that round receives an EMA update. The student uses SGD with momentum 0.9, weight decay 10^{-4} and the registered polynomial learning-rate decay. All floating BN statistics and integer counters are saved. Supervision is pixel CE plus a ramped consistency loss, without the additional image BCE used by the DINO dense systems. Image scores aggregate the top 5% crack-versus-other log-odds. Equal mask identities therefore do not imply identical training objectives or compute.

Weight-averaged teacher targets, adversarial prediction smoothness and rectangular image mixing build on Mean Teacher [20], VAT [22] and CutMix [21], respectively. Our port applies VAT in encoder-feature space through two frozen teacher decoders, and mixes spatial pseudo-logits and validity masks with image regions. Its Beta(4,4) setting differs from the original CutMix experiments' Beta(1,1). These are documented adaptations of component principles; original classification or medical-segmentation scores are not matched EL results.

The 69-fit protocol comprises 18 source hyperparameter fits, three fixed literal-VAT-setting sensitivity fits, 36 SCDD development fits, and 12 matched UCF20% source fits. The supervised architecture uses 20% and 100% training masks; SSL uses 20% plus the remaining source-training images without reading their masks. DINO frozen, LoRA and normal-reference source controls use the same 86 training masks and 144 validation masks. Full-source UCF and upstream dense hyperparameter calibration are additional annotation exposure, not hidden within the 20% claim.

The port uses canonical three-class labels, whole-field 336-pixel letterboxing, batch two on MPS, synchronized D4 views and explicitly counted updates. These differ from the original five-class, crop-based experiment. The primary VAT finite-difference scale is 0.1, the released helper default. An initial local FP32 source-only precision probe found a substantially diminished perturbation with the released caller's scale 10^{-6}; that literal setting remains a fixed three-seed source sensitivity. This diagnostic does not demonstrate failure of the authors' trained system on other hardware.

Primary inference uses teacher 1, matching released PV-S3 inference. A predeclared mean of both teacher logits, reflecting PS-MT paper inference, is evaluated at the identical source-selected checkpoint with its own source-calibrated operating point. It is a correlated inference view, not another trained seed or an option chosen using target scores. Exact FP32 parameters and complete normalization state are stored losslessly; full-resolution CPU forward/backward and fresh-model recovery checks passed with zero source-prediction error in the implementation probe. Those probe scores are excluded from research tables. All 18 source-calibration fits and the complete 69-fit full-method/matched-control schedule are complete and audited. These adaptations are not a reproduction of the published reported score.

B Endpoint definitions and complete reporting support

Image decisions use the strict inequality $s_i > \tau_{\text{source}}$ on the original saved score. Recall and FPR use annotated positive and negative image denominators; review is the fraction of all images flagged. AUROC and AP retain their different prevalence and ranking meanings. Source temperature is never fitted on target outcomes.

For known native pixels, the crack map uses the fixed strict score rule $p > 0.5$; the source-calibrated distance system retains its model-specific response convention. With crack truth G_i and predicted crack pixels P_i, Dice is $2|G_i \cap P_i|/(|G_i| + |P_i|)$ and IoU is $|G_i \cap P_i|/|G_i \cup P_i|$. Positive-image means exclude images without crack truth. All-image Dice, which can reward empty true-negative masks, does not replace positive-image overlap. Native pooled AP concatenates known pixels within each bounded SCDD/UCF cohort and is a whole-cohort point estimate. It is neither mean per-image AP nor an independent-pixel bootstrap.

SCDD normal-structure FPR pools false-positive pixels over annotated normal-structure pixels in crack-negative images only (the original raw annotation classes 4 and 9 after canonical eligibility). UCF lacks that annotation and remains null. A negative-field mean pixel FPR is a separate descriptive endpoint. PVEL box-point hit tests whether the maximum-response pixel centre falls in any target box, with the first row-major pixel resolving exact ties. It does not measure object-detection AP or mask Dice.

All 1,683 prediction jobs passed the original execution audit and the separate saved-output checks. The 630 spatial jobs comprise 496 SCDD/UCF native-mask jobs and 134 PVEL box jobs. Five fixed native-failure strata for each eligible job yield 2,480 records; 116 have empty denominators and retain null rates. The three positive bins are

$(0, 0.5\%]$, $(0.5\%, 2\%]$ and above 2% of known-pixel crack annotation. Zero overlap and detected-with-zero-overlap are literal events; a misplaced nonempty mask can have zero overlap. No physical crack severity is inferred from annotation area.

The 3,780 fixed display maps comprise 2,172 SCDD, 804 UCF and 804 PVEL maps, grouped into 216 local sheets. The original metadata-only cases are retained, including weak responses. These float16 display products do not replace full-precision metric summaries. Raw-image/mask redistribution is not inferred from local access.

All 311 PVEL views use the once-scored 12,326-image pool. The primary 12,292-image population, full pool and broader 11,927-image sensitivity are derived from the same saved scores; a line-only analysis additionally excludes star-crack positives under its fixed rule. The original 17 near-copy pairs and all 34 excluded members remain documented. Neither integrity screening nor these sensitivities establishes physical module identity, crop independence or pretraining exclusion. There is no PVEL capture-family confidence interval.

Calibration companions retain 5,049 model/cohort/metric rows for Brier score, negative log-likelihood and fixed-bin ECE. These diagnostics use the original whole pools of SCDD117, UCF507 and PVEL12,326, not the matching held-source strata or primary PVEL population. The four diagnostic sheets additionally preserve all native groups and all 3,732 population-sensitivity rows. Their descriptive points are correlated through fitted systems and image reuse.

C Interpretation of all original control questions

Every Q-number below follows the order of the original pre-scoring question rationale. The trace table preserves each question's full FIRST/SECOND family IDs, cohort, stratum, endpoint, original mean and every fitted-pair difference. Question membership overlaps, so 6,252 memberships refer to 6,212 unique declarations. Each conditional interval remains unadjusted across this large family. Positive signed differences are not automatically beneficial, particularly for FPR and review burden.

C.1 Q1: Adaptation across original image controls

Most signed AUROC differences favor the original second-family image LoRA systems, but adverse and tied conditions remain. This does not establish a universally improved inspection operating point. Frozen features, full fine-tuning and CNN comparators have different pretraining and optimization histories; budgets and acquisition strata overlap.

C.2 Q2: Frozen DINO and classical features with the same classifier

The declared frozen-feature comparisons mostly have positive AUROC differences, with an adverse condition retained. The result concerns complete feature/preprocessing systems. Sharing a classifier does not match representation pretraining, dimensionality, preprocessing or annotation-dependent HPO.

C.3 Q3: LoRA and classical controls

All declared AUROC mean differences are positive in the original orientation. Thresholded review and transferred PVEL behavior require their own records. Neural adaptation and classical tuning have unequal search and optimization budgets; the comparisons do not isolate representation or training cost.

C.4 Q4: Adaptation relative to frozen/classical predictors

All declared AUROC mean differences are positive, conditional on these source-selected systems and fixed cohorts. The controls overlap the other image themes and are not independent confirmations of the same claim.

C.5 Q5: Frozen representation systems

Signed AUROC differences are predominantly positive but include an adverse comparison. Retain each original feature and processing recipe. Pretraining, geometry, dimensionality and pooling are confounded; an encoder architecture cause is not identified.

C.6 Q6: Processing scale

AUROC differences have both signs. Increasing processing scale is not uniformly beneficial across the declared source and target conditions. Resizing changes the complete input processing system; it does not recover missing physical information or establish a resolution mechanism.

C.7 Q7: CLS and patch pooling

Both signed AUROC directions occur; pooling effects depend on processing size and source condition. CLS, patch mean and concatenation differ in feature geometry and dimensionality. No pooling is selected from confirmation outcomes.

C.8 Q8: Frozen larger encoder

The original larger-encoder comparisons have positive AUROC differences, while positive PVEL recall changes accompany positive author-normal FPR changes. Larger models and pretraining differ. Recall alone does not establish improved operating decisions or resource efficiency.

C.9 Q9: Model scale

Both signed AUROC directions occur, although positive means are more frequent. The full original model-scale conditions remain separate. Parameter count and pretraining are not isolated causes; shared fitted systems and cohorts are correlated.

C.10 Q10: ResNet and DINO with a common classifier

All declared AUROC mean differences are positive in the original orientation. ImageNet and DINO pretraining differ. This is a complete frozen-feature comparison, not a randomized architecture comparison.

C.11 Q11: Dense adaptation and original same-mask controls

AUROC differences are mostly positive, whereas native Dice has nearly balanced positive and adverse means. Image ranking and native overlap do not provide interchangeable evidence. Same masks do not match pretraining, losses, optimization or source model selection. Keep all fitted-pair directions and pooled-AP point estimates.

C.12 Q12: Normal reference relative to frozen dense prediction

AUROC and Dice contain positive and adverse conditions; positive native-AP means do not imply uniform positive-image overlap or review improvements. Memory coverage, retrieval, decoder fitting and source annotation structure remain alternative explanations; no matching-specific cause is established.

C.13 Q13: Local patch objective and decoder

AUROC and Dice both have mixed signed differences. Changing objective and decoder does not produce an endpoint-uniform gain. These are complete systems with different supervision pathways and decoder structure, despite matching original annotation IDs where verified.

C.14 Q14: Adapted multidepth and local patch systems

The frozen declaration is adapted multidepth LoRA FIRST and patch MLP SECOND. All fourteen AUROC differences are negative; eleven of fourteen Dice mean differences are negative. The direction must not be reversed to match the prose label. Encoder adaptation, feature depth and decoder objectives change together.

C.15 Q15: Source-calibrated normal distance

The original normal-distance SECOND comparisons have adverse AUROC and native-AP differences and often larger normal-structure FPR. Their complete source cutoff is retained. A source normal-distance cutoff is not softmax 0.5 or a crack posterior. Matching and supervised decoder changes cannot be separated.

C.16 Q16: Fresh decoder after encoder adaptation

Fresh-head AUROC and Dice means include positive, adverse and tied behavior; extra head optimization can change the complete system. Fresh decoder initialization, encoder freezing and prior adaptation are jointly changed; parent work must be charged separately.

C.17 Q17: Reference within matched fresh decoders

Reference versus no-reference fresh heads have both signed AUROC and Dice differences, including ties. There is no uniform reference benefit. Same trainable count removes one difference but does not identify a unique retrieval mechanism or erase seed sensitivity.

C.18 Q18: Full post-adaptation system and original parent

Both positive and adverse AUROC and Dice effects remain, while native AP includes adverse means. Parent and new decoder stages are distinct fitted components. Additional optimization and changed encoder/decoder roles preclude a reference-only causal interpretation.

C.19 Q19: Full supervised local published-architecture controls

Many original SECOND dense-comparator means improve AUROC and Dice, while native AP retains numerous adverse conditions. The matched UCF LoRA gain transfers with lower mean SCDD Dice. Published-port objectives, pretraining and optimization work differ. These are local ports, not reproductions or rankings of published headline metrics.

C.20 Q20: Semi-supervised and supervised local ports

AUROC has both signed directions and Dice includes adverse and tied means. Unlabelled exposure does not guarantee a native-overlap benefit. The semi-supervised objective and unlabelled images are additional inputs. Teacher views share fitted weights and do not create independent training replications.

C.21 Q21: Dense controls and semi-supervised local ports

All declared AUROC means are positive in the original orientation, but native AP has more adverse than positive means. Dice and operating decisions need their own original records. Source-selected semi-supervised systems, primary teacher and ensemble are correlated; pretraining and optimizer work differ.

C.22 Q22: Teacher ensemble

Original teacher-ensemble differences have both AUROC directions and positive, adverse or tied Dice. Ensembling is not uniformly beneficial. The views share training. Their differences describe inference choices frozen before scoring, not independent fitted-model replications.

C.23 Q23: Geometry and local evidence aggregation

Position and top-5%-versus-mean controls have mixed AUROC and Dice differences. No geometry or pooling setting is selected after scoring. The frozen controls differ in explicit geometry or aggregation; small cohorts and fitted-seed sensitivity limit mechanistic inference.

C.24 Q24: Stronger image CNN optimization controls

All twelve original LoRA-minus-additional-CNN AUROC means and all thirty-six fitted-pair directions are positive; the strongest ResNet-50 SCDD interval still includes zero. The CNN rates were post-hoc source sensitivities but fixed before confirmation. These reused fits and two cohorts are not twelve independent studies or an architecture effect.

C.25 Q25: Literal numerical port sensitivity

All four original SECOND literal-recipe AUROC, Dice and native-AP mean differences are negative. The selected-port and literal-port failures remain visible. The complete numerical recipe changes more than xi; this is not an xi-only causal test. No target-driven repair is made.

C.26 Q26: Literal-port correlated ensemble

The two AUROC differences have opposite directions, while both Dice means increase. The ensemble and teacher share fitting and the literal numerical recipe. Positive Dice does not imply a general ensemble benefit.

C.27 Q27: Matched UCF20% DINO controls

Frozen-to-LoRA and reference-to-LoRA AUROC and Dice means are positive in both SCDD117 and UCF507. These four DINO contrasts do not denote the supervised-port comparison. The 86 training and 144 validation masks are nested within 574 development/HPO images; matched labels do not match pretraining, HPO or optimization.

C.28 Q28: Matched UCF20% normal reference

Both original frozen-to-reference AUROC and Dice mean differences are negative on SCDD117 and UCF507. This is evidence against a universal reference benefit in these fits, not proof that reference information cannot help under other fixed systems.

C.29 Q29: Original UCF-only reference transfer

AUROC, Dice and native AP each retain positive and adverse future-transfer conditions. Source-only fitted weights do not make the target dataset unseen to research selection. Both datasets informed study-wide design, and transfer remains conditional on source-selected systems.

D Complete fixed dense-control numerical tables

All six pre-existing presentation matrix scopes; all 456 original comparisons, without outcome selection. Raw ratios are not percentages. Fitted-seed signs follow original pairs, including deterministic reuse; they are not additional training replication. No PVEL acquisition-family interval or pooled-native-AP interval is invented.

Table S1: SCDD ARTS primary stratum, 43 images. Complete fixed dense-control scope 1. Raw endpoint ratios; differences are SECOND minus FIRST. Intervals are conditional and unadjusted. Codes resolve in the appended recipe legend.

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F001→F009	AUROC	0.5558	0.8071	+0.2512	[+0.1374, +0.3386]	+,+,+
F001→F009	Image AP	0.4086	0.761	+0.3525	[+0.1729, +0.4873]	+,+,+
F001→F009	Recall	0.5714	0.7381	+0.1667	[+0.02222, +0.2963]	-,+,-
F001→F009	Review	0.5891	0.4884	−0.1008	[−0.2302, +0.01639]	-,+,-
F001→F009	Image FPR	0.5977	0.3678	−0.2299	[−0.3601, −0.09674]	-,+,-
F001→F009	Pos. Dice	0.1014	0.0921	−0.009334	[−0.05713, +0.01539]	-,+,-
F001→F009	Pos. IoU	0.06023	0.05072	−0.00951	[−0.04166, +0.006503]	-,+,-
F001→F009	Structure FPR	0.0004425	0.0001291	−0.0003134	[−0.0008627, +0.0002021]	-,-,-
F008→F009	AUROC	0.5616	0.8071	+0.2455	[+0.1277, +0.3403]	+,+,+
F008→F009	Image AP	0.4074	0.761	+0.3536	[+0.2007, +0.4784]	+,+,+
F008→F009	Recall	0.5476	0.7381	+0.1905	[+0.02564, +0.3333]	+,+,+
F008→F009	Review	0.4729	0.4884	+0.0155	[−0.09912, +0.1333]	-,+,+
F008→F009	Image FPR	0.4368	0.3678	−0.06897	[−0.1976, +0.06667]	-,0,0
F008→F009	Pos. Dice	0.05873	0.0921	+0.03337	[+0.01076, +0.05589]	+,+,-
F008→F009	Pos. IoU	0.03186	0.05072	+0.01886	[+0.006647, +0.03155]	+,+,+
F008→F009	Structure FPR	0.0001175	0.0001291	$+1.152 \times 10^{-5}$	$[0, +4.404 \times 10^{-5}]$	+,+,-
F010→F009	AUROC	0.5821	0.8071	+0.225	[+0.09923, +0.3372]	+,+,+
F010→F009	Image AP	0.4219	0.761	+0.3392	[+0.1803, +0.4573]	+,+,+
F010→F009	Recall	0.4524	0.7381	+0.2857	[+0.03326, +0.5112]	+,+,+
F010→F009	Review	0.3953	0.4884	+0.09302	[−0.04577, +0.2315]	-,+,+
F010→F009	Image FPR	0.3678	0.3678	$+1.85 \times 10^{-17}$	[−0.1415, +0.1333]	-,+,0
F010→F009	Pos. Dice	0.06304	0.0921	+0.02906	[+0.004592, +0.0508]	+,+,+
F010→F009	Pos. IoU	0.03397	0.05072	+0.01674	[+0.003474, +0.02875]	+,+,+
F010→F009	Structure FPR	0.0002305	0.0001291	−0.0001014	[−0.0002976, 0]	+,+,-
F039→F047	AUROC	0.6158	0.8571	+0.2414	[+0.06447, +0.3697]	+,+,+
F039→F047	Image AP	0.4735	0.827	+0.3535	[+0.1824, +0.4869]	+,+,+
F039→F047	Recall	0.9524	0.9524	0	[−0.1667, +0.1111]	0,0,0
F039→F047	Review	0.9147	0.814	−0.1008	[−0.2222, +0.01633]	+,-,-
F039→F047	Image FPR	0.8966	0.7471	−0.1494	[−0.2949, +0.02861]	+,-,-
F039→F047	Pos. Dice	0.1339	0.1015	−0.03249	[−0.08328, +0.002292]	+,-,-
F039→F047	Pos. IoU	0.07534	0.05635	−0.01899	[−0.04834, +0.000643]	+,-,-
F039→F047	Structure FPR	0.00366	0.0003803	−0.003279	[−0.005957, −0.001036]	-,-,-
F046→F047	AUROC	0.6174	0.8571	+0.2397	[+0.05594, +0.3969]	+,+,+
F046→F047	Image AP	0.5308	0.827	+0.2961	[+0.05202, +0.4617]	+,+,+
F046→F047	Recall	0.7619	0.9524	+0.1905	[+0.05556, +0.3751]	+,+,+
F046→F047	Review	0.6977	0.814	+0.1163	[+0.01708, +0.2248]	+,+,+
F046→F047	Image FPR	0.6667	0.7471	+0.08046	[−0.01906, +0.193]	+,+,+
F046→F047	Pos. Dice	0.09069	0.1015	+0.01077	[−0.01707, +0.05545]	+,+,-
F046→F047	Pos. IoU	0.04935	0.05635	+0.007003	[−0.009865, +0.03434]	+,+,-
F046→F047	Structure FPR	0.0003146	0.0003803	$+6.568 \times 10^{-5}$	[−0.0001419, +0.0002804]	-,+,+
F048→F047	AUROC	0.6314	0.8571	+0.2258	[+0.04252, +0.3732]	+,+,+
F048→F047	Image AP	0.5558	0.827	+0.2712	[+0.03139, +0.4391]	+,+,+
F048→F047	Recall	0.7381	0.9524	+0.2143	[0, +0.4048]	+,+,+
F048→F047	Review	0.6512	0.814	+0.1628	[+0.04346, +0.2857]	+,+,+
F048→F047	Image FPR	0.6092	0.7471	+0.1379	[+0.01332, +0.2821]	+,+,-
F048→F047	Pos. Dice	0.1056	0.1015	−0.004125	[−0.02423, +0.01489]	+,-,+
F048→F047	Pos. IoU	0.0579	0.05635	−0.001548	[−0.01357, +0.01049]	+,-,+
F048→F047	Structure FPR	0.000257	0.0003803	+0.0001233	$[-7.378 \times 10^{-5}, +0.0002828]$	+,+,+

Table S2: SCDD CFVS primary stratum, 27 images. Complete fixed dense-control scope 2. Raw endpoint ratios; differences are SECOND minus FIRST. Intervals are conditional and unadjusted. Codes resolve in the appended recipe legend.

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F075→F083	AUROC	0.5926	1	+0.4074	[+0.2844, +0.5429]	+,+,+
F075→F083	Image AP	0.6997	1	+0.3003	[+0.1419, +0.4903]	+,+,+
F075→F083	Recall	1	1	0	[0, 0]	0,0,0
F075→F083	Review	0.9753	0.8765	−0.09877	[−0.1667, −0.03922]	0,-,-
F075→F083	Image FPR	0.9444	0.7222	−0.2222	[−0.3056, −0.1111]	0,-,-
F075→F083	Pos. Dice	0.138	0.2052	+0.06717	[+0.01522, +0.1234]	+,-,+
F075→F083	Pos. IoU	0.08331	0.1183	+0.03504	[+0.000398, +0.07206]	+,-,+
F075→F083	Structure FPR	2.616×10^{-5}	0	-2.616×10^{-5}	$[-9.339 \times 10^{-5}, 0]$	-,0,0
F082→F083	AUROC	0.8815	1	+0.1185	[+0.03332, +0.1958]	+,+,+
F082→F083	Image AP	0.9338	1	+0.06617	[+0.02273, +0.1155]	+,+,+
F082→F083	Recall	0.9111	1	+0.08889	[+0.02564, +0.1515]	0,+,+
F082→F083	Review	0.7531	0.8765	+0.1235	[+0.04598, +0.2133]	0,+,+
F082→F083	Image FPR	0.5556	0.7222	+0.1667	[0, +0.3111]	0,0,+
F082→F083	Pos. Dice	0.1597	0.2052	+0.04545	[+0.0234, +0.06815]	+,+,+
F082→F083	Pos. IoU	0.09083	0.1183	+0.02752	[+0.01344, +0.04184]	+,+,+
F082→F083	Structure FPR	0.003132	0	−0.003132	[−0.008145, 0]	-,0,-
F084→F083	AUROC	0.7093	1	+0.2907	[+0.1747, +0.4173]	+,+,+
F084→F083	Image AP	0.7763	1	+0.2237	[+0.1228, +0.3365]	+,+,+
F084→F083	Recall	0.7778	1	+0.2222	[+0.1111, +0.3095]	0,+,+
F084→F083	Review	0.642	0.8765	+0.2346	[+0.1449, +0.3189]	0,+,+
F084→F083	Image FPR	0.4722	0.7222	+0.25	[+0.09086, +0.381]	0,0,+
F084→F083	Pos. Dice	0.08089	0.2052	+0.1243	[+0.08532, +0.1673]	+,+,+
F084→F083	Pos. IoU	0.04487	0.1183	+0.07348	[+0.0489, +0.101]	+,+,+
F084→F083	Structure FPR	0.003222	0	−0.003222	[−0.008918, 0]	-,0,-
F113→F121	AUROC	0.8944	1	+0.1056	[+0.001316, +0.2377]	+,+,+
F113→F121	Image AP	0.9469	1	+0.05306	[+0.0008553, +0.132]	+,+,+
F113→F121	Recall	0.8667	1	+0.1333	[0, +0.313]	+,+,+
F113→F121	Review	0.5802	0.5556	−0.02469	[−0.1528, +0.1061]	-,+,-
F113→F121	Image FPR	0.2222	0	−0.2222	[−0.3333, −0.1026]	-,0,-
F113→F121	Pos. Dice	0.2312	0.2053	−0.02596	[−0.08119, +0.02635]	-,+,-
F113→F121	Pos. IoU	0.1355	0.1167	−0.01888	[−0.05302, +0.01343]	-,-,-
F113→F121	Structure FPR	5.231×10^{-6}	0	-5.231×10^{-6}	$[-1.563 \times 10^{-5}, 0]$	-,0,0
F120→F121	AUROC	0.8833	1	+0.1167	[+0.04936, +0.1845]	+,+,+
F120→F121	Image AP	0.9286	1	+0.07135	[+0.02653, +0.136]	+,+,+
F120→F121	Recall	0.8222	1	+0.1778	[+0.03326, +0.2963]	+,+,+
F120→F121	Review	0.5556	0.5556	$+5.551 \times 10^{-17}$	[−0.12, +0.125]	+,-,+
F120→F121	Image FPR	0.2222	0	−0.2222	[−0.303, −0.1333]	0,-,0
F120→F121	Pos. Dice	0.07488	0.2053	+0.1304	[+0.1158, +0.1484]	+,+,+
F120→F121	Pos. IoU	0.04132	0.1167	+0.07534	[+0.06576, +0.08708]	+,+,+
F120→F121	Structure FPR	1.569×10^{-5}	0	-1.569×10^{-5}	$[-5.603 \times 10^{-5}, 0]$	-,0,0
F122→F121	AUROC	0.9593	1	+0.04074	[0, +0.09584]	+,+,+
F122→F121	Image AP	0.9759	1	+0.02411	[0, +0.05822]	+,+,+
F122→F121	Recall	0.9111	1	+0.08889	[0, +0.1587]	+,+,0
F122→F121	Review	0.6296	0.5556	−0.07407	[−0.1975, +0.03226]	-,+,-
F122→F121	Image FPR	0.2778	0	−0.2778	[−0.4334, −0.1333]	-,0,-
F122→F121	Pos. Dice	0.05097	0.2053	+0.1543	[+0.1334, +0.181]	+,+,+
F122→F121	Pos. IoU	0.02868	0.1167	+0.08797	[+0.0751, +0.1038]	+,+,+
F122→F121	Structure FPR	0.001833	0	−0.001833	[−0.003601, −0.0001465]	-,-,0

Table S3: SCDD CSIR primary stratum, 47 images. Complete fixed dense-control scope 3. Raw endpoint ratios; differences are SECOND minus FIRST. Intervals are conditional and unadjusted. Codes resolve in the appended recipe legend.

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F149→F157	AUROC	0.878	0.9318	+0.05374	[−0.02942, +0.1695]	+,+,+
F149→F157	Image AP	0.8982	0.9499	+0.05169	[−0.006442, +0.2226]	+,+,+
F149→F157	Recall	0.9583	0.9583	-3.701×10^{-17}	[−0.04762, +0.03924]	-,0,+
F149→F157	Review	0.695	0.8014	+0.1064	[+0.03245, +0.2]	-,+,+
F149→F157	Image FPR	0.4203	0.6377	+0.2174	[+0.08772, +0.3333]	0,+,+
F149→F157	Pos. Dice	0.2728	0.1793	−0.09356	[−0.198, +0.04196]	-,-,-
F149→F157	Pos. IoU	0.1786	0.1045	−0.07413	[−0.1469, +0.01908]	-,-,-
F149→F157	Structure FPR	0	0.0002444	+0.0002444	[0, +0.0008031]	+,0,+
F156→F157	AUROC	0.6558	0.9318	+0.276	[+0.1443, +0.4313]	+,+,+
F156→F157	Image AP	0.62	0.9499	+0.3299	[+0.1285, +0.5322]	+,+,+
F156→F157	Recall	0.9861	0.9583	−0.02778	[−0.1001, 0]	-,-,0
F156→F157	Review	0.9433	0.8014	−0.1418	[−0.2456, −0.0705]	-,-,-
F156→F157	Image FPR	0.8986	0.6377	−0.2609	[−0.3958, −0.1373]	-,-,-
F156→F157	Pos. Dice	0.2017	0.1793	−0.02248	[−0.05644, +0.02207]	+,-,+
F156→F157	Pos. IoU	0.1186	0.1045	−0.01405	[−0.03537, +0.0132]	+,-,+
F156→F157	Structure FPR	0.0008782	0.0002444	−0.0006338	[−0.001586, +0.0001323]	-,-,-
F158→F157	AUROC	0.6292	0.9318	+0.3025	[+0.167, +0.4652]	+,+,+
F158→F157	Image AP	0.6002	0.9499	+0.3497	[+0.1371, +0.5724]	+,+,+
F158→F157	Recall	0.9861	0.9583	−0.02778	[−0.1001, 0]	-,-,0
F158→F157	Review	0.9433	0.8014	−0.1418	[−0.2456, −0.0705]	-,-,-
F158→F157	Image FPR	0.8986	0.6377	−0.2609	[−0.3958, −0.1373]	-,-,-
F158→F157	Pos. Dice	0.1925	0.1793	−0.01322	[−0.03627, +0.02399]	+,-,+
F158→F157	Pos. IoU	0.1117	0.1045	−0.007174	[−0.02144, +0.01641]	+,-,+
F158→F157	Structure FPR	0.001298	0.0002444	−0.001054	[−0.003218, +0.0003607]	-,-,-
F187→F195	AUROC	0.6238	0.9595	+0.3357	[+0.1684, +0.551]	+,+,+
F187→F195	Image AP	0.6537	0.9698	+0.316	[+0.1424, +0.5865]	+,+,+
F187→F195	Recall	0.8056	0.9722	+0.1667	[−0.04171, +0.4494]	+,+,+
F187→F195	Review	0.7234	0.7305	+0.007092	[−0.1712, +0.1879]	+,-,+
F187→F195	Image FPR	0.6377	0.4783	−0.1594	[−0.3651, +0.07415]	-,-,-
F187→F195	Pos. Dice	0.1571	0.1894	+0.03231	[−0.03597, +0.1201]	+,+,+
F187→F195	Pos. IoU	0.09371	0.1083	+0.01464	[−0.02572, +0.06835]	+,-,+
F187→F195	Structure FPR	0.001433	0.00115	−0.0002825	[−0.002244, +0.002239]	+,-,+
F194→F195	AUROC	0.7808	0.9595	+0.1787	[+0.06844, +0.3135]	+,+,+
F194→F195	Image AP	0.746	0.9698	+0.2237	[+0.05593, +0.4601]	+,+,+
F194→F195	Recall	1	0.9722	−0.02778	[−0.1001, 0]	0,-,-
F194→F195	Review	0.8511	0.7305	−0.1206	[−0.2222, −0.05128]	-,-,-
F194→F195	Image FPR	0.6957	0.4783	−0.2174	[−0.359, −0.08333]	-,-,-
F194→F195	Pos. Dice	0.211	0.1894	−0.02164	[−0.03704, −0.01367]	+,-,-
F194→F195	Pos. IoU	0.1218	0.1083	−0.01343	[−0.02242, −0.008422]	+,-,-
F194→F195	Structure FPR	0.003282	0.00115	−0.002131	$[-0.004779, -9.676 \times 10^{-5}]$	-,-,-
F196→F195	AUROC	0.7766	0.9595	+0.183	[+0.07307, +0.3205]	+,+,+
F196→F195	Image AP	0.7467	0.9698	+0.223	[+0.06168, +0.462]	+,+,+
F196→F195	Recall	1	0.9722	−0.02778	[−0.1001, 0]	0,-,-
F196→F195	Review	0.8794	0.7305	−0.1489	[−0.2564, −0.06802]	-,-,-
F196→F195	Image FPR	0.7536	0.4783	−0.2754	[−0.4127, −0.1282]	-,-,-
F196→F195	Pos. Dice	0.1886	0.1894	+0.0007719	[−0.01595, +0.01249]	+,-,-
F196→F195	Pos. IoU	0.107	0.1083	+0.001343	[−0.008669, +0.008655]	+,-,-
F196→F195	Structure FPR	0.002826	0.00115	−0.001676	[−0.00382, −0.0001142]	-,+,-

Table S4: SCDD whole primary stratum, 117 images. Complete fixed dense-control scope 4. Raw endpoint ratios; differences are SECOND minus FIRST. Intervals are conditional and unadjusted. Codes resolve in the appended recipe legend.

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F223→F232	AUROC	0.583	0.7952	+0.2122	[+0.1152, +0.2994]	+,+,+
F223→F232	Image AP	0.5687	0.758	+0.1893	[+0.09595, +0.2659]	+,+,+
F223→F232	Recall	0.9623	0.9434	−0.01887	[−0.07972, +0.03774]	-,-,+
F223→F232	Review	0.9487	0.812	−0.1368	[−0.19, −0.0862]	-,-,-
F223→F232	Image FPR	0.9375	0.7031	−0.2344	[−0.2925, −0.1695]	-,-,-
F223→F232	Pos. Dice	0.05237	0.1446	+0.09222	[+0.06752, +0.1149]	+,+,+
F223→F232	Pos. IoU	0.03054	0.08274	+0.0522	[+0.03761, +0.06566]	+,+,+
F223→F232	Structure FPR	0.00112	0.003826	+0.002706	[+0.0006854, +0.004697]	+,0,+
F223→F232	Native AP	0.1374	0.1513	+0.01393	point only	-,+,+
F230→F232	AUROC	0.6327	0.7952	+0.1625	[+0.09706, +0.2243]	+,+,+
F230→F232	Image AP	0.601	0.758	+0.157	[+0.07707, +0.2179]	+,+,+
F230→F232	Recall	0.9182	0.9434	+0.02516	[−0.01212, +0.07407]	+,-,+
F230→F232	Review	0.8604	0.812	−0.04843	[−0.09041, −0.006112]	-,-,+
F230→F232	Image FPR	0.8125	0.7031	−0.1094	[−0.1667, −0.04877]	-,-,+
F230→F232	Pos. Dice	0.06801	0.1446	+0.07659	[+0.0591, +0.09472]	+,+,+
F230→F232	Pos. IoU	0.03859	0.08274	+0.04415	[+0.03321, +0.05512]	+,+,+
F230→F232	Structure FPR	0.00292	0.003826	+0.0009064	[−0.003799, +0.004646]	+,-,-
F230→F232	Native AP	0.1257	0.1513	+0.02567	point only	+,-,+
F231→F232	AUROC	0.7275	0.7952	+0.06771	[+0.03717, +0.09639]	+,+,-
F231→F232	Image AP	0.6541	0.758	+0.1039	[+0.05526, +0.1324]	+,+,-
F231→F232	Recall	0.9497	0.9434	−0.006289	[−0.0303, +0.01587]	0,-,0
F231→F232	Review	0.8746	0.812	−0.06268	[−0.1006, −0.02823]	-,-,-
F231→F232	Image FPR	0.8125	0.7031	−0.1094	[−0.1667, −0.05085]	-,-,-
F231→F232	Pos. Dice	0.1259	0.1446	+0.0187	[+0.002018, +0.03816]	+,-,+
F231→F232	Pos. IoU	0.07374	0.08274	+0.009001	[−0.0007989, +0.02025]	+,-,+
F231→F232	Structure FPR	0.000628	0.003826	+0.003198	[+0.0015, +0.005262]	+,-,+
F231→F232	Native AP	0.1419	0.1513	+0.009485	point only	-,+,+
F233→F232	AUROC	0.6828	0.7952	+0.1124	[+0.07141, +0.1504]	+,+,+
F233→F232	Image AP	0.6366	0.758	+0.1214	[+0.05692, +0.1683]	+,+,+
F233→F232	Recall	0.8994	0.9434	+0.04403	[+0.006402, +0.09032]	+,0,+
F233→F232	Review	0.8348	0.812	−0.02279	[−0.06377, +0.02013]	-,-,+
F233→F232	Image FPR	0.7812	0.7031	−0.07812	[−0.1368, −0.01754]	-,-,+
F233→F232	Pos. Dice	0.1467	0.1446	−0.002144	[−0.01375, +0.01132]	-,-,+
F233→F232	Pos. IoU	0.08421	0.08274	−0.001467	[−0.008509, +0.006872]	-,-,+
F233→F232	Structure FPR	0.002309	0.003826	+0.001517	[−0.0007402, +0.003946]	+,-,-
F233→F232	Native AP	0.147	0.1513	+0.004359	point only	-,+,+
F259→F268	AUROC	0.7094	0.8917	+0.1823	[+0.1039, +0.253]	+,+,+
F259→F268	Image AP	0.729	0.894	+0.165	[+0.09283, +0.2414]	+,+,+
F259→F268	Recall	0.9371	0.956	+0.01887	[−0.03335, +0.06672]	-,-,+
F259→F268	Review	0.8661	0.7749	−0.09117	[−0.1534, −0.0352]	-,-,+
F259→F268	Image FPR	0.8073	0.625	−0.1823	[−0.2752, −0.09722]	-,-,+
F259→F268	Pos. Dice	0.1959	0.129	−0.06693	[−0.1069, −0.02463]	-,+,-
F259→F268	Pos. IoU	0.1199	0.0754	−0.04448	[−0.07063, −0.0172]	-,+,-
F259→F268	Structure FPR	0.0004636	0.0001824	−0.0002812	[−0.0008179, +0.000136]	-,-,-
F259→F268	Native AP	0.2238	0.1756	−0.04812	point only	-,+,-
F266→F268	AUROC	0.682	0.8917	+0.2097	[+0.148, +0.2649]	+,+,+
F266→F268	Image AP	0.6739	0.894	+0.22	[+0.1297, +0.3079]	+,+,+
F266→F268	Recall	0.956	0.956	0	[−0.03333, +0.03154]	-,-,+
F266→F268	Review	0.8832	0.7749	−0.1083	[−0.1634, −0.05469]	-,-,+
F266→F268	Image FPR	0.8229	0.625	−0.1979	[−0.2752, −0.118]	-,-,-
F266→F268	Pos. Dice	0.09141	0.129	+0.03759	[+0.02137, +0.0556]	+,+,-
F266→F268	Pos. IoU	0.05187	0.0754	+0.02353	[+0.01365, +0.03466]	+,+,-
F266→F268	Structure FPR	0.001669	0.0001824	−0.001487	[−0.002634, −0.0005747]	+,-,-
F266→F268	Native AP	0.1466	0.1756	+0.02899	point only	+,+,-
F267→F268	AUROC	0.8698	0.8917	+0.02191	[−0.00812, +0.05071]	+,+,-
F267→F268	Image AP	0.873	0.894	+0.02099	[−0.003353, +0.05148]	+,+,+
F267→F268	Recall	0.9811	0.956	−0.02516	[−0.06219, +0.006667]	-,-,+
F267→F268	Review	0.849	0.7749	−0.07407	[−0.1186, −0.03405]	-,-,+

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F267→F268	Image FPR	0.7396	0.625	−0.1146	[−0.1841, −0.0476]	-,-,+
F267→F268	Pos. Dice	0.1336	0.129	−0.004605	[−0.02229, +0.01733]	+,+,-
F267→F268	Pos. IoU	0.07728	0.0754	−0.001879	[−0.01157, +0.01033]	+,+,-
F267→F268	Structure FPR	0.0001511	0.0001824	$+3.134 \times 10^{-5}$	[−0.0002032, +0.0003426]	+,-,0
F267→F268	Native AP	0.1679	0.1756	+0.007734	point only	+,+,-
F269→F268	AUROC	0.7503	0.8917	+0.1414	[+0.09188, +0.1909]	+,+,+
F269→F268	Image AP	0.7482	0.894	+0.1458	[+0.08203, +0.2278]	+,+,+
F269→F268	Recall	0.9119	0.956	+0.04403	[+0.006803, +0.08773]	+,-,+
F269→F268	Review	0.8433	0.7749	−0.06838	[−0.1275, −0.01232]	-,-,+
F269→F268	Image FPR	0.7865	0.625	−0.1615	[−0.25, −0.07812]	-,-,-
F269→F268	Pos. Dice	0.1104	0.129	+0.01857	[+0.005647, +0.03179]	+,+,+
F269→F268	Pos. IoU	0.06283	0.0754	+0.01258	[+0.004418, +0.02118]	+,+,+
F269→F268	Structure FPR	0.000544	0.0001824	−0.0003616	$[-0.0007807, -3.463 \times 10^{-5}]$	-,-,-
F269→F268	Native AP	0.1637	0.1756	+0.01194	point only	+,+,-
F295→F304	AUROC	0.7774	0.9293	+0.1519	[+0.05576, +0.254]	+,+,+
F295→F304	Image AP	0.7964	0.9285	+0.1321	[+0.04958, +0.246]	+,+,+
F295→F304	Recall	0.9057	0.9371	+0.03145	[−0.06384, +0.1163]	-,+,+
F295→F304	Review	0.812	0.6382	−0.1738	[−0.2549, −0.09524]	-,-,-
F295→F304	Image FPR	0.7344	0.3906	−0.3438	[−0.4167, −0.2567]	-,-,-
F295→F304	Pos. Dice	0.1631	0.1689	+0.005892	[−0.02641, +0.03848]	+,-,-
F295→F304	Pos. IoU	0.09637	0.09704	+0.0006718	[−0.01874, +0.02083]	+,-,-
F295→F304	Structure FPR	0.0008743	0.0007919	-8.236×10^{-5}	[−0.0006618, +0.0005957]	+,-,+
F295→F304	Native AP	0.1977	0.1624	−0.03534	point only	-,-,-
F302→F304	AUROC	0.763	0.9293	+0.1664	[+0.0921, +0.2664]	+,+,+
F302→F304	Image AP	0.7618	0.9285	+0.1667	[+0.07696, +0.3071]	+,+,+
F302→F304	Recall	0.9308	0.9371	+0.006289	[−0.0513, +0.05883]	0,+,0
F302→F304	Review	0.7892	0.6382	−0.151	[−0.2245, −0.08608]	-,-,-
F302→F304	Image FPR	0.6719	0.3906	−0.2812	[−0.375, −0.1937]	-,-,-
F302→F304	Pos. Dice	0.1098	0.1689	+0.05911	[+0.03894, +0.07941]	+,+,+
F302→F304	Pos. IoU	0.06262	0.09704	+0.03442	[+0.02238, +0.04655]	+,+,+
F302→F304	Structure FPR	0.00104	0.0007919	−0.0002479	[−0.001407, +0.0005868]	-,+,+
F302→F304	Native AP	0.1359	0.1624	+0.02643	point only	-,+,+
F303→F304	AUROC	0.926	0.9293	+0.003341	[−0.02328, +0.02845]	+,-,-
F303→F304	Image AP	0.9325	0.9285	−0.004025	[−0.02906, +0.01667]	-,+,-
F303→F304	Recall	0.9623	0.9371	−0.02516	[−0.07346, +0.0101]	-,-,-
F303→F304	Review	0.735	0.6382	−0.09687	[−0.1554, −0.04478]	-,-,-
F303→F304	Image FPR	0.5469	0.3906	−0.1562	[−0.2537, −0.06667]	-,0,-
F303→F304	Pos. Dice	0.1707	0.1689	−0.001778	[−0.01785, +0.0152]	+,-,-
F303→F304	Pos. IoU	0.09886	0.09704	−0.001824	[−0.01138, +0.008285]	+,-,-
F303→F304	Structure FPR	8.437×10^{-5}	0.0007919	+0.0007075	[+0.0001957, +0.00143]	+,+,+
F303→F304	Native AP	0.1722	0.1624	−0.009832	point only	-,-,-
F305→F304	AUROC	0.764	0.9293	+0.1654	[+0.08774, +0.2625]	+,+,+
F305→F304	Image AP	0.7572	0.9285	+0.1713	[+0.07803, +0.3017]	+,+,+
F305→F304	Recall	0.9434	0.9371	−0.006289	[−0.0667, +0.04631]	-,0,0
F305→F304	Review	0.8091	0.6382	−0.1709	[−0.2456, −0.1034]	-,-,-
F305→F304	Image FPR	0.6979	0.3906	−0.3073	[−0.399, −0.2154]	-,-,-
F305→F304	Pos. Dice	0.145	0.1689	+0.02395	[+0.009964, +0.04092]	-,+,+
F305→F304	Pos. IoU	0.0818	0.09704	+0.01524	[+0.006636, +0.0259]	-,+,+
F305→F304	Structure FPR	0.001383	0.0007919	−0.0005914	[−0.00158, +0.0002523]	-,-,-
F305→F304	Native AP	0.1656	0.1624	−0.003237	point only	-,+,-
F336→F345	AUROC	0.7708	0.9631	+0.1923	[+0.1055, +0.2975]	+,+,+
F336→F345	Image AP	0.7927	0.9614	+0.1687	[+0.08109, +0.3111]	+,+,+
F336→F345	Recall	0.8491	0.9748	+0.1258	[+0.03844, +0.2258]	+,+,+
F336→F345	Review	0.698	0.6496	−0.04843	[−0.1382, +0.03253]	+,-,-
F336→F345	Image FPR	0.5729	0.3802	−0.1927	[−0.3153, −0.08695]	-,-,-
F336→F345	Pos. Dice	0.1814	0.1536	−0.02775	[−0.06905, +0.01812]	-,+,-
F336→F345	Pos. IoU	0.1065	0.08844	−0.01801	[−0.04377, +0.009991]	-,-,-
F336→F345	Structure FPR	0.000769	3.134×10^{-5}	−0.0007377	[−0.001615, −0.000109]	-,-,-
F336→F345	Native AP	0.216	0.1677	−0.04829	point only	-,-,-
F343→F345	AUROC	0.8404	0.9631	+0.1227	[+0.05404, +0.2114]	+,+,+
F343→F345	Image AP	0.8146	0.9614	+0.1468	[+0.05129, +0.2986]	+,+,+
F343→F345	Recall	0.8994	0.9748	+0.07547	[+0.02345, +0.1429]	+,+,+

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F343→F345	Review	0.641	0.6496	+0.008547	$[-0.0697, +0.08951]$	+,-,+
F343→F345	Image FPR	0.4271	0.3802	−0.04688	$[-0.179, +0.08461]$	+,-,-
F343→F345	Pos. Dice	0.1437	0.1536	+0.009905	$[-0.01007, +0.02998]$	-,+,+
F343→F345	Pos. IoU	0.08281	0.08844	+0.00563	$[-0.006265, +0.01778]$	-,+,+
F343→F345	Structure FPR	0.001156	3.134×10^{-5}	−0.001125	$[-0.002428, -0.0001681]$	-,+,-
F343→F345	Native AP	0.1619	0.1677	+0.005845	point only	-,+,+
F344→F345	AUROC	0.9576	0.9631	+0.005503	$[-0.00937, +0.02361]$	+,-,+
F344→F345	Image AP	0.9559	0.9614	+0.005504	$[-0.009589, +0.02538]$	+,-,+
F344→F345	Recall	0.9748	0.9748	0	$[-0.025, +0.02667]$	+,0,-
F344→F345	Review	0.6781	0.6496	−0.02849	$[-0.07921, +0.01575]$	+,-,-
F344→F345	Image FPR	0.4323	0.3802	−0.05208	$[-0.1378, +0.02605]$	+,-,-
F344→F345	Pos. Dice	0.1514	0.1536	+0.002255	$[-0.01483, +0.02061]$	-,+,+
F344→F345	Pos. IoU	0.08638	0.08844	+0.00206	$[-0.008362, +0.01332]$	-,+,+
F344→F345	Structure FPR	5.424×10^{-5}	3.134×10^{-5}	-2.29×10^{-5}	$[-7.859 \times 10^{-5}, +1.455 \times 10^{-5}]$	-,+,+
F344→F345	Native AP	0.164	0.1677	+0.003791	point only	-,+,+
F346→F345	AUROC	0.8362	0.9631	+0.127	$[+0.05952, +0.2181]$	+,+,+
F346→F345	Image AP	0.8162	0.9614	+0.1452	$[+0.05278, +0.2943]$	+,+,+
F346→F345	Recall	0.8931	0.9748	+0.08176	$[+0.0227, +0.1543]$	+,+,+
F346→F345	Review	0.6496	0.6496	0	$[-0.08048, +0.08039]$	+,-,-
F346→F345	Image FPR	0.4479	0.3802	−0.06771	$[-0.2029, +0.06768]$	+,-,-
F346→F345	Pos. Dice	0.1535	0.1536	+0.000132	$[-0.02063, +0.02247]$	-,+,+
F346→F345	Pos. IoU	0.08826	0.08844	+0.0001844	$[-0.01247, +0.01318]$	-,+,+
F346→F345	Structure FPR	0.0004765	3.134×10^{-5}	−0.0004452	$[-0.0009261, -9.488 \times 10^{-5}]$	-,-,-
F346→F345	Native AP	0.17	0.1677	−0.002306	point only	-,+,+
F389→F398	AUROC	0.7015	0.9114	+0.2099	$[+0.1264, +0.3111]$	+,+,+
F389→F398	Image AP	0.7063	0.9242	+0.218	$[+0.1097, +0.3392]$	+,+,+
F389→F398	Recall	1	0.9937	−0.006289	$[-0.02128, 0]$	0,0,-
F389→F398	Review	1	0.906	−0.09402	$[-0.1323, -0.06018]$	-,-,-
F389→F398	Image FPR	1	0.8333	−0.1667	$[-0.2188, -0.1146]$	-,-,-
F389→F398	Pos. Dice	0.06667	0.06642	−0.0002443	$[-0.0173, +0.01608]$	+,+,-
F389→F398	Pos. IoU	0.03589	0.03593	$+3.646 \times 10^{-5}$	$[-0.009623, +0.00899]$	+,+,-
F389→F398	Structure FPR	0.09408	0.001043	−0.09304	$[-0.119, -0.06944]$	-,-,-
F389→F398	Native AP	0.1364	0.1005	−0.0359	point only	-,-,-
F396→F398	AUROC	0.8049	0.9114	+0.1064	$[+0.04619, +0.1907]$	+,+,+
F396→F398	Image AP	0.8266	0.9242	+0.09764	$[+0.03619, +0.2033]$	+,+,+
F396→F398	Recall	0.9371	0.9937	+0.0566	$[-0.006536, +0.1418]$	+,+,+
F396→F398	Review	0.8689	0.906	+0.03704	$[-0.02425, +0.09919]$	+,+,-
F396→F398	Image FPR	0.8125	0.8333	+0.02083	$[-0.06154, +0.1061]$	+,+,-
F396→F398	Pos. Dice	0.08244	0.06642	−0.01602	$[-0.02911, -0.003878]$	+,+,-
F396→F398	Pos. IoU	0.04459	0.03593	−0.008665	$[-0.01584, -0.001873]$	+,+,-
F396→F398	Structure FPR	0.03817	0.001043	−0.03713	$[-0.06063, -0.01804]$	-,-,-
F396→F398	Native AP	0.102	0.1005	−0.001512	point only	+,-,-
F397→F398	AUROC	0.8702	0.9114	+0.04118	$[+0.01163, +0.08104]$	+,+,+
F397→F398	Image AP	0.8917	0.9242	+0.03256	$[+0.008229, +0.07051]$	+,+,+
F397→F398	Recall	0.9686	0.9937	+0.02516	$[+0.004898, +0.05131]$	+,0,-
F397→F398	Review	0.9145	0.906	−0.008547	$[-0.04721, +0.02609]$	+,-,-
F397→F398	Image FPR	0.8698	0.8333	−0.03646	$[-0.09942, +0.02083]$	+,-,-
F397→F398	Pos. Dice	0.103	0.06642	−0.03658	$[-0.05124, -0.02152]$	+,-,-
F397→F398	Pos. IoU	0.05673	0.03593	−0.0208	$[-0.02915, -0.01223]$	+,-,-
F397→F398	Structure FPR	0.01131	0.001043	−0.01027	$[-0.02075, -0.002758]$	-,-,-
F397→F398	Native AP	0.1336	0.1005	−0.03318	point only	+,-,-
F427→F398	AUROC	0.7976	0.9114	+0.1138	$[+0.05462, +0.2003]$	+,+,+
F427→F398	Image AP	0.8215	0.9242	+0.1028	$[+0.04083, +0.2119]$	+,+,+
F427→F398	Recall	0.9371	0.9937	+0.0566	$[0, +0.1277]$	+,+,+
F427→F398	Review	0.8775	0.906	+0.02849	$[-0.02521, +0.08477]$	+,+,-
F427→F398	Image FPR	0.8281	0.8333	+0.005208	$[-0.06966, +0.08219]$	+,+,-
F427→F398	Pos. Dice	0.07895	0.06642	−0.01252	$[-0.02268, -0.001117]$	+,+,-
F427→F398	Pos. IoU	0.04279	0.03593	−0.006863	$[-0.01238, -0.0006646]$	+,+,-
F427→F398	Structure FPR	0.04166	0.001043	−0.04061	$[-0.06414, -0.02297]$	-,-,-
F427→F398	Native AP	0.1072	0.1005	−0.006733	point only	+,+,-

Table S5: PVEL primary known-crack/author-normal cohort, 12,292 images. Complete fixed dense-control scope 5. Raw endpoint ratios; differences are SECOND minus FIRST. Intervals are conditional and unadjusted. Codes resolve in the appended recipe legend.

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F295→F304	Known recall	1	0.9965	−0.003535	point only	0,-,-
F295→F304	Normal FPR	1	0.9812	−0.01883	point only	0,-,-
F295→F304	Box-point hit	0.2467	0.3627	+0.1159	point only	+,+,+
F302→F304	Known recall	0.9972	0.9965	−0.000707	point only	0,-,+
F302→F304	Normal FPR	0.9905	0.9812	−0.009369	point only	0,-,+
F302→F304	Box-point hit	0.257	0.3627	+0.1057	point only	+,+,+
F303→F304	Known recall	1	0.9965	−0.003535	point only	0,-,-
F303→F304	Normal FPR	1	0.9812	−0.01883	point only	0,-,-
F303→F304	Box-point hit	0.234	0.3627	+0.1287	point only	+,+,+
F305→F304	Known recall	0.9915	0.9965	+0.004949	point only	0,-,+
F305→F304	Normal FPR	0.9746	0.9812	+0.00655	point only	0,-,+
F305→F304	Box-point hit	0.3666	0.3627	−0.003888	point only	+,+,-
F336→F345	Known recall	0.9996	0.9519	−0.04772	point only	0,-,-
F336→F345	Normal FPR	0.9999	0.8538	−0.1462	point only	-,-,-
F336→F345	Box-point hit	0.5419	0.3588	−0.1831	point only	-,-,+
F343→F345	Known recall	1	0.9519	−0.04807	point only	-,-,-
F343→F345	Normal FPR	0.9987	0.8538	−0.1449	point only	-,-,-
F343→F345	Box-point hit	0.2563	0.3588	+0.1025	point only	+,+,+
F344→F345	Known recall	1	0.9519	−0.04807	point only	-,-,-
F344→F345	Normal FPR	1	0.8538	−0.1462	point only	-,-,-
F344→F345	Box-point hit	0.4136	0.3588	−0.05479	point only	-,+,-
F346→F345	Known recall	0.9993	0.9519	−0.04737	point only	-,-,-
F346→F345	Normal FPR	0.9975	0.8538	−0.1437	point only	-,-,-
F346→F345	Box-point hit	0.2344	0.3588	+0.1244	point only	+,+,+
F389→F398	Known recall	1	0.9452	−0.05479	point only	-,-,-
F389→F398	Normal FPR	0.9994	0.8476	−0.1519	point only	-,-,-
F389→F398	Box-point hit	0.3446	0.2474	−0.09721	point only	-,+,-
F396→F398	Known recall	0.9555	0.9452	−0.01025	point only	+,-,+
F396→F398	Normal FPR	0.9035	0.8476	−0.05595	point only	+,-,+
F396→F398	Box-point hit	0.2739	0.2474	−0.02651	point only	+,+,-
F397→F398	Known recall	0.9173	0.9452	+0.02793	point only	+,-,+
F397→F398	Normal FPR	0.7889	0.8476	+0.05862	point only	+,-,+
F397→F398	Box-point hit	0.2743	0.2474	−0.02686	point only	+,-,-
F427→F398	Known recall	0.9753	0.9452	−0.03005	point only	-,-,-
F427→F398	Normal FPR	0.9384	0.8476	−0.09085	point only	-,-,+
F427→F398	Box-point hit	0.2467	0.2474	+0.000707	point only	+,+,-

Table S6: UCF primary cohort, 507 images. Complete fixed dense-control scope 6. Raw endpoint ratios; differences are SECOND minus FIRST. Intervals are conditional and unadjusted. Codes resolve in the appended recipe legend.

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F295→F304	AUROC	0.7032	0.9314	+0.2282	[+0.1757, +0.2875]	+,+,+
F295→F304	Image AP	0.5621	0.9334	+0.3714	[+0.2884, +0.4665]	+,+,+
F295→F304	Recall	0.7238	0.9466	+0.2228	[+0.152, +0.2929]	+,+,-
F295→F304	Review	0.62	0.6667	+0.04668	[−0.02989, +0.1237]	+,+,-
F295→F304	Image FPR	0.5624	0.5112	−0.05112	[−0.1437, +0.04103]	+,+,-
F295→F304	Pos. Dice	0.01719	0.2662	+0.249	[+0.2118, +0.283]	+,+,+
F295→F304	Pos. IoU	0.009863	0.1818	+0.1719	[+0.142, +0.2004]	+,+,+
F295→F304	Native AP	0.1033	0.3852	+0.2819	point only	+,+,+
F302→F304	AUROC	0.8805	0.9314	+0.05086	[+0.02297, +0.08193]	+,+,+
F302→F304	Image AP	0.8515	0.9334	+0.08193	[+0.04585, +0.1417]	+,+,+
F302→F304	Recall	0.9724	0.9466	−0.02578	[−0.06092, +0.002359]	-,+,-
F302→F304	Review	0.8573	0.6667	−0.1907	[−0.2583, −0.1217]	-,+,-

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F302→F304	Image FPR	0.7935	0.5112	−0.2822	[−0.3684, −0.1842]	-,0,-
F302→F304	Pos. Dice	0.2156	0.2662	+0.05065	[+0.03448, +0.06922]	-,+,+
F302→F304	Pos. IoU	0.1382	0.1818	+0.04356	[+0.02978, +0.05851]	-,+,+
F302→F304	Native AP	0.3248	0.3852	+0.06039	point only	-,+,+
F303→F304	AUROC	0.9206	0.9314	+0.01078	[+0.0007557, +0.02334]	+,+,+
F303→F304	Image AP	0.9252	0.9334	+0.008235	[−0.001075, +0.01977]	+,+,+
F303→F304	Recall	0.9613	0.9466	−0.01473	[−0.02948, −0.001802]	-,+,-
F303→F304	Review	0.8041	0.6667	−0.1374	[−0.1787, −0.1035]	-,+,-
F303→F304	Image FPR	0.7168	0.5112	−0.2055	[−0.2568, −0.1606]	-,+,-
F303→F304	Pos. Dice	0.1992	0.2662	+0.06698	[+0.05004, +0.08258]	-,+,+
F303→F304	Pos. IoU	0.1267	0.1818	+0.05502	[+0.04301, +0.06649]	-,+,+
F303→F304	Native AP	0.3652	0.3852	+0.01993	point only	-,+,+
F305→F304	AUROC	0.8859	0.9314	+0.04545	[+0.02108, +0.07326]	+,+,+
F305→F304	Image AP	0.8687	0.9334	+0.06473	[+0.03493, +0.1121]	+,+,+
F305→F304	Recall	0.9429	0.9466	+0.003683	[−0.02128, +0.02969]	-,+,+
F305→F304	Review	0.7659	0.6667	−0.09928	[−0.1463, −0.05169]	-,-,+
F305→F304	Image FPR	0.6677	0.5112	−0.1564	[−0.2192, −0.08934]	-,-,+
F305→F304	Pos. Dice	0.2634	0.2662	+0.002848	[−0.02308, +0.03279]	-,+,+
F305→F304	Pos. IoU	0.1721	0.1818	+0.009646	[−0.00849, +0.03055]	-,+,+
F305→F304	Native AP	0.341	0.3852	+0.04415	point only	-,+,+
F336→F345	AUROC	0.6924	0.9347	+0.2422	[+0.1899, +0.2943]	+,+,+
F336→F345	Image AP	0.6092	0.9362	+0.327	[+0.2486, +0.4233]	+,+,+
F336→F345	Recall	0.9576	0.8932	−0.06446	[−0.1228, −0.01598]	-,0,-
F336→F345	Review	0.8961	0.4181	−0.478	[−0.5577, −0.4023]	-,-,-
F336→F345	Image FPR	0.862	0.1544	−0.7076	[−0.76, −0.6469]	-,-,-
F336→F345	Pos. Dice	0.05754	0.2186	+0.161	[+0.09651, +0.224]	+,+,+
F336→F345	Pos. IoU	0.03495	0.148	+0.113	[+0.06294, +0.1636]	+,+,+
F336→F345	Native AP	0.1239	0.4922	+0.3682	point only	+,+,+
F343→F345	AUROC	0.927	0.9347	+0.00766	[−0.01181, +0.02992]	+,+,-
F343→F345	Image AP	0.9077	0.9362	+0.02856	[+0.008513, +0.06612]	+,+,+
F343→F345	Recall	0.9595	0.8932	−0.0663	[−0.1174, −0.02594]	-,-,-
F343→F345	Review	0.6706	0.4181	−0.2525	[−0.3189, −0.1918]	-,-,-
F343→F345	Image FPR	0.5102	0.1544	−0.3558	[−0.4361, −0.2798]	-,-,-
F343→F345	Pos. Dice	0.2244	0.2186	−0.005833	[−0.03806, +0.02244]	-,+,-
F343→F345	Pos. IoU	0.1511	0.148	−0.003109	[−0.02654, +0.01837]	-,+,-
F343→F345	Native AP	0.3767	0.4922	+0.1155	point only	+,+,+
F344→F345	AUROC	0.9401	0.9347	−0.005446	[−0.01879, +0.007647]	-,-,-
F344→F345	Image AP	0.9406	0.9362	−0.004352	[−0.01659, +0.008351]	-,+,+
F344→F345	Recall	0.9908	0.8932	−0.09761	[−0.1565, −0.05468]	-,-,-
F344→F345	Review	0.908	0.4181	−0.4898	[−0.5706, −0.4155]	-,-,-
F344→F345	Image FPR	0.862	0.1544	−0.7076	[−0.7599, −0.6474]	-,-,-
F344→F345	Pos. Dice	0.2962	0.2186	−0.07763	[−0.1153, −0.04437]	-,-,-
F344→F345	Pos. IoU	0.2026	0.148	−0.05466	[−0.08039, −0.03106]	-,-,-
F344→F345	Native AP	0.4782	0.4922	+0.01394	point only	+,-,+
F346→F345	AUROC	0.9306	0.9347	+0.004022	[−0.01475, +0.02687]	+,+,-
F346→F345	Image AP	0.9159	0.9362	+0.02036	[+0.001983, +0.05435]	+,+,+
F346→F345	Recall	0.9669	0.8932	−0.07366	[−0.1222, −0.0359]	-,-,-
F346→F345	Review	0.7015	0.4181	−0.2834	[−0.3451, −0.2256]	-,-,-
F346→F345	Image FPR	0.5542	0.1544	−0.3998	[−0.4714, −0.3324]	-,-,-
F346→F345	Pos. Dice	0.2669	0.2186	−0.04832	[−0.08426, −0.01387]	-,-,-
F346→F345	Pos. IoU	0.1796	0.148	−0.03162	[−0.05775, −0.006203]	-,-,-
F346→F345	Native AP	0.4035	0.4922	+0.08867	point only	+,+,+
F389→F398	AUROC	0.9511	0.9616	+0.0106	[−0.004246, +0.02641]	+,+,-
F389→F398	Image AP	0.9422	0.954	+0.01175	[−0.002788, +0.02975]	+,+,-
F389→F398	Recall	0.9816	0.954	−0.02762	[−0.05476, −0.005416]	0,-,-
F389→F398	Review	0.7061	0.5457	−0.1604	[−0.2082, −0.1148]	-,-,-
F389→F398	Image FPR	0.5532	0.319	−0.2342	[−0.2948, −0.175]	-,-,-
F389→F398	Pos. Dice	0.492	0.4723	−0.01971	[−0.04982, +0.01505]	+,+,-
F389→F398	Pos. IoU	0.371	0.3668	−0.004221	[−0.03186, +0.0297]	+,+,-
F389→F398	Native AP	0.6652	0.6352	−0.03002	point only	+,+,-
F396→F398	AUROC	0.9458	0.9616	+0.0159	[+0.003163, +0.03223]	+,+,+
F396→F398	Image AP	0.9352	0.954	+0.01872	[+0.006362, +0.04169]	+,+,+

FIRST → SECOND	Endpoint	FIRST	SECOND	Difference	Interval	Pair signs
F396→F398	Recall	0.954	0.954	$+3.701 \times 10^{-17}$	$[-0.02345, +0.02716]$	+,-,-
F396→F398	Review	0.5812	0.5457	-0.0355	$[-0.09035, +0.01574]$	+,-,-
F396→F398	Image FPR	0.3742	0.319	-0.05521	$[-0.1386, +0.02314]$	+,-,-
F396→F398	Pos. Dice	0.5301	0.4723	-0.05778	$[-0.08451, -0.03496]$	+,+,-
F396→F398	Pos. IoU	0.4087	0.3668	-0.04198	$[-0.06132, -0.0249]$	+,+,-
F396→F398	Native AP	0.5837	0.6352	$+0.05143$	point only	+,+,+
F397→F398	AUROC	0.9634	0.9616	-0.001779	$[-0.007396, +0.003535]$	+,+,-
F397→F398	Image AP	0.9567	0.954	-0.002781	$[-0.009826, +0.002669]$	+,+,-
F397→F398	Recall	0.9392	0.954	$+0.01473$	$[-0.001756, +0.03676]$	+,-,-
F397→F398	Review	0.4753	0.5457	$+0.07035$	$[+0.04065, +0.09745]$	+,-,+
F397→F398	Image FPR	0.2178	0.319	$+0.1012$	$[+0.06027, +0.1367]$	+,-,+
F397→F398	Pos. Dice	0.5447	0.4723	-0.07241	$[-0.1004, -0.04904]$	+,-,-
F397→F398	Pos. IoU	0.4294	0.3668	-0.06262	$[-0.08649, -0.04134]$	-,-,-
F397→F398	Native AP	0.7033	0.6352	-0.06813	point only	+,-,-
F427→F398	AUROC	0.9395	0.9616	$+0.02219$	$[+0.005957, +0.04178]$	+,+,+
F427→F398	Image AP	0.9301	0.954	$+0.02388$	$[+0.009398, +0.05063]$	+,+,+
F427→F398	Recall	0.9466	0.954	$+0.007366$	$[-0.01678, +0.03779]$	+,-,-
F427→F398	Review	0.6029	0.5457	-0.0572	$[-0.1168, -0.002198]$	+,-,-
F427→F398	Image FPR	0.4121	0.319	-0.09305	$[-0.1831, -0.007518]$	+,-,-
F427→F398	Pos. Dice	0.5043	0.4723	-0.03202	$[-0.05314, -0.01136]$	+,+,-
F427→F398	Pos. IoU	0.3877	0.3668	-0.02098	$[-0.03654, -0.006507]$	+,+,-
F427→F398	Native AP	0.5875	0.6352	$+0.04771$	point only	+,+,-

Recipe codes for every table above

Code	Original recipe, source unit, budget and inference view
F001	scdd / holdout_ARTS / b0p2 / deeplabv3plus_full / primary
F009	scdd / holdout_ARTS / b0p2 / dinov3_dense_lora / primary
F008	scdd / holdout_ARTS / b0p2 / dinov3_dense_frozen / primary
F010	scdd / holdout_ARTS / b0p2 / dinov3_dense_reference / primary
F039	scdd / holdout_ARTS / b1 / deeplabv3plus_full / primary
F047	scdd / holdout_ARTS / b1 / dinov3_dense_lora / primary
F046	scdd / holdout_ARTS / b1 / dinov3_dense_frozen / primary
F048	scdd / holdout_ARTS / b1 / dinov3_dense_reference / primary
F075	scdd / holdout_CFVS / b0p2 / deeplabv3plus_full / primary
F083	scdd / holdout_CFVS / b0p2 / dinov3_dense_lora / primary
F082	scdd / holdout_CFVS / b0p2 / dinov3_dense_frozen / primary
F084	scdd / holdout_CFVS / b0p2 / dinov3_dense_reference / primary
F113	scdd / holdout_CFVS / b1 / deeplabv3plus_full / primary
F121	scdd / holdout_CFVS / b1 / dinov3_dense_lora / primary
F120	scdd / holdout_CFVS / b1 / dinov3_dense_frozen / primary
F122	scdd / holdout_CFVS / b1 / dinov3_dense_reference / primary
F149	scdd / holdout_CSIR / b0p2 / deeplabv3plus_full / primary
F157	scdd / holdout_CSIR / b0p2 / dinov3_dense_lora / primary
F156	scdd / holdout_CSIR / b0p2 / dinov3_dense_frozen / primary
F158	scdd / holdout_CSIR / b0p2 / dinov3_dense_reference / primary
F187	scdd / holdout_CSIR / b1 / deeplabv3plus_full / primary
F195	scdd / holdout_CSIR / b1 / dinov3_dense_lora / primary
F194	scdd / holdout_CSIR / b1 / dinov3_dense_frozen / primary
F196	scdd / holdout_CSIR / b1 / dinov3_dense_reference / primary
F223	scdd / internal / b0p05 / deeplabv3plus_full / primary
F232	scdd / internal / b0p05 / dinov3_dense_lora / primary
F230	scdd / internal / b0p05 / dinov3_dense_frozen / primary
F231	scdd / internal / b0p05 / dinov3_dense_full / primary
F233	scdd / internal / b0p05 / dinov3_dense_reference / primary
F259	scdd / internal / b0p1 / deeplabv3plus_full / primary
F268	scdd / internal / b0p1 / dinov3_dense_lora / primary
F266	scdd / internal / b0p1 / dinov3_dense_frozen / primary
F267	scdd / internal / b0p1 / dinov3_dense_full / primary
F269	scdd / internal / b0p1 / dinov3_dense_reference / primary
F295	scdd / internal / b0p2 / deeplabv3plus_full / primary
F304	scdd / internal / b0p2 / dinov3_dense_lora / primary
F302	scdd / internal / b0p2 / dinov3_dense_frozen / primary
F303	scdd / internal / b0p2 / dinov3_dense_full / primary
F305	scdd / internal / b0p2 / dinov3_dense_reference / primary
F336	scdd / internal / b1 / deeplabv3plus_full / primary
F345	scdd / internal / b1 / dinov3_dense_lora / primary
F343	scdd / internal / b1 / dinov3_dense_frozen / primary
F344	scdd / internal / b1 / dinov3_dense_full / primary
F346	scdd / internal / b1 / dinov3_dense_reference / primary
F389	ucf / source / b1 / deeplabv3plus_full / primary
F398	ucf / source / b1 / dinov3_dense_lora / primary
F396	ucf / source / b1 / dinov3_dense_frozen / primary
F397	ucf / source / b1 / dinov3_dense_full / primary
F427	ucf / source / b1 / ucf_reference_position / primary

E Source context, computational cost and the separate ELPV supplement

All original source schedules completed: 36 neural calibration and 180 development fits; 70 classical calibration and 180 development records; 168 dense fits; nine UCF-reference fits; 18 stronger-CNN stability fits; 379 representation records; and 125 mechanism records. The complete published-method schedule contains 69 audited source/matched fits, including 60 full-architecture and nine matched DINO fits. These counts refer to different catalogue stages, shared views and deterministic reuse; they are not a sum of independent architectures or all upstream HPO work.

The original full-budget internal dense source comparison has LoRA Dice 0.149 versus adapted CNN 0.198 despite higher image AUROC. The stronger source image-only ResNet-50 AUROC of 0.969 exceeds image-only LoRA 0.956. Matched UCF20% source LoRA Dice is 0.404889 versus 0.307231 for the full supervised local port, with favorable directions in all three fits, while native AP and review behavior are not uniformly better. These source observations preceded confirmation and remain distinguished from the main article's actual confirmation values.

Formal efficiency covers 97 fitted cases, 388 cohort/batch conditions and 1,164 measured whole-cohort passes on the 32 GiB M2 Max. It uses 351 SCDD and 430 UCF source-training images, batches one and eight, warm-ups and three uninstrumented measured passes, with separate instrumentation and loading observations. Full-budget dense LoRA averages 31.73 ms/image on SCDD at batch one versus CNN 27.08, with mixed fitted-pair directions. RSS lifetime highwater and sampled MPS allocations overlap and are never added. These 97 cases do not independently time every final checkpoint.

The ELPV supplement uses only 1,968 course-development images and four grouped outer folds. Its 24 completed outer fits follow 480 inner-grid fits, with fold-specific scaling and parameter selection. Frozen DINOv3 CLS plus RBF SVM has mean outer AUROC 0.904918 versus 0.838261 for HOG/statistics plus RBF SVM; all four fold differences are positive. Overlapping training folds and expert-review labels prevent treating this as independent crack confirmation, a CNN comparison or a training-population interval. The 656 course-test images remain unscored. All six systems, eight endpoints and fold values remain in the fourteen-page vector source/cost/ELPV numerical companion.

F Reproduction boundaries, provenance and retained failures

The owner-side records bind 26,259 authorized original inputs and a completed output closure of 11,142 saved job files, including 1,683 result/prediction pairs, 3,780 fixed maps and 630 original cache-retirement receipts. Full inventories and image identities are owner-only. The frozen scientific closure contains 92 exact files and one already observed registry. A separately authorized reproduction must satisfy the original path and byte guards; missing inputs cannot be replaced with symlinks, random weights, fabricated labels or skipped oracles.

The code package retains the sibling research/course code layout, dependency pins, model initialization and local-port provenance. The exact four-author-file recipe specifies the original PS-MT encoder, ResNet implementation, MIT notice and PV-S3 loss oracle by pinned source and hash. Upstream PV-S3 oracle source has no verified redistribution licence and is excluded from the transferable code-only package; an authorized recipient must review terms and obtain those exact inputs. The PV-S3 numerical oracle check remains mandatory and must fail explicitly when its inputs are absent. The owner's completed acquisition probe does not grant redistribution rights.

Historical implementation evidence consists of 171 original-environment and 164 separate fresh-environment checks with 53 pins on the same host/base Python. These are synthetic, metadata and numerical-oracle checks, not OS isolation, cross-device determinism, fresh-training performance reproduction or tests of every later reporting script. Complete saved-output auditing and the separately registered same-host statistical replay have separate evidence and scope. The replay completed successfully and reproduced every saved-job and declared contrast field apart from generation time; the original analysis bytes remained unchanged. This is same-host, same-environment, same-implementation repeatability on the same saved outputs, not independent-implementation, cross-device, OS-isolation or fresh-training reproduction.

Original training and reporting failures remain append-only. The zero-extra-update recovery retains its original 19-round/950-step fit, optimizer/RNG and selected weights. Three owner-pause-contaminated round/fit timers and a recovery-finalization timer remain raw; they are not silently converted to active compute. The candidate-inventory v1 seed-grouping failure and separate v2 grouping-only amendment retain all 1,061 original

run/weight/threshold/source-output fields. Reporting-only JSON parsing, omitted output lists, ambiguous recipe labels and clipped single-panel figures have explicit retained originals and presentation supersession records. None changes the frozen evaluator, scientific populations, scores or choices.

Raw EL images, masks, labels, fitted/pretrained weights, optimizer/RNG payloads, prediction arrays and full owner input/output inventories are excluded from the sanitized private handoff. Dataset, model and code terms remain distinct. No public release, external submission or contact is performed. Author and venue fields remain intentionally deferred by the owner.

Companion navigation and historical record

The numerical delivery set contains: (i) all 6,212 ordered contrasts with intervals and every fitted-pair value; (ii) 3,686 absolute family/endpoint cells and 9,292 fitted-member endpoint values; (iii) the all-29-question trace and direction/null counts; (iv) all calibration, native-failure and population-sensitivity rows; (v) the complete forest atlas and six dense matrices; and (vi) fourteen source/cost/ELPV vector sheets with numerical companions. Complete recipe ledgers retain all 1,061 views and 427 families. Fixed spatial sheets remain owner-local pending image-specific distribution permission. No companion selects a best target model or a new hypothesis.

The entire earlier development/protocol narrative and all embedded assets remain byte-preserved below in this same editable source under the default-off `includehistoricalrecord` switch, as well as in immutable revision archives and earlier exported PDFs. Historical progress statements belong to that earlier record. They are not current scientific conclusions or missing confirmation analyses.